

Differential pair rules in gastrointestinal aGVHD

What the method resolves, what agrees with the atlas, and what is new

What is measured. For every ordered pair of cell types and every FOV, the mining step asks whether the two types sit closer together than chance (attraction), further apart than chance (avoidance), or neither. A field of view (FOV) is *eligible* for a rule only when it holds at least 20 cells of both named types. Among eligible FOVs the rule is attraction, avoidance, or no rule, and the net score is $100(N_{\text{attraction}} - N_{\text{avoidance}})/N_{\text{eligible}}$. Direction matters: $A \rightarrow B$ and $B \rightarrow A$ are separate rules with their own counts and their own tests. The run behind this document holds 525 directional pair rules over 267 FOVs.

Which severity score. Stages are **clinical** unless a figure says otherwise, because that is the score the atlas paper uses for its own immune analyses. The pathological score is shown wherever the two disagree, and [they disagree a great deal](#) – which turns out to be one of the findings rather than a nuisance.

How to read the tests. “Ordered FDR” asks whether the net score moves steadily from Control to Mild to Severe. C–M, M–S and C–S compare the two named stages. FDR is corrected across all rules in that screen; below 0.05 is called significant. Rule names are never chosen by hand before a screen.

Which trends count. A result is reported here only when Control and Severe differ. A rule that jumps at Mild and returns to its Control value by Severe is not treated as a finding, however small its Mild–Severe FDR; those stay in the notebooks.

What this cannot say. These are spatial associations, not mechanisms, between classifier labels. Many endpoint groups are small. Every figure prints its own denominators, and those should be read before its FDR.

1 Does the method see real spatial structure?

One question comes before any biology: does a pair rule measure how cells are *arranged*, or is it a slow way of counting how many cells there are? Nothing in this part depends on a rule name chosen in advance.

The screen returns a stable, structured ranking

A few pair rules occur in most fields of view, the rest fall away smoothly, and the stage columns already separate. Paneth $\rightarrow$ Epithelial drops from 21% of Control FOVs to 8% of clinically Severe ones; Endocrine $\rightarrow$ Epithelial climbs from 4% to 21%; CD4T $\rightarrow$ Plasma falls from 38% to 3%.

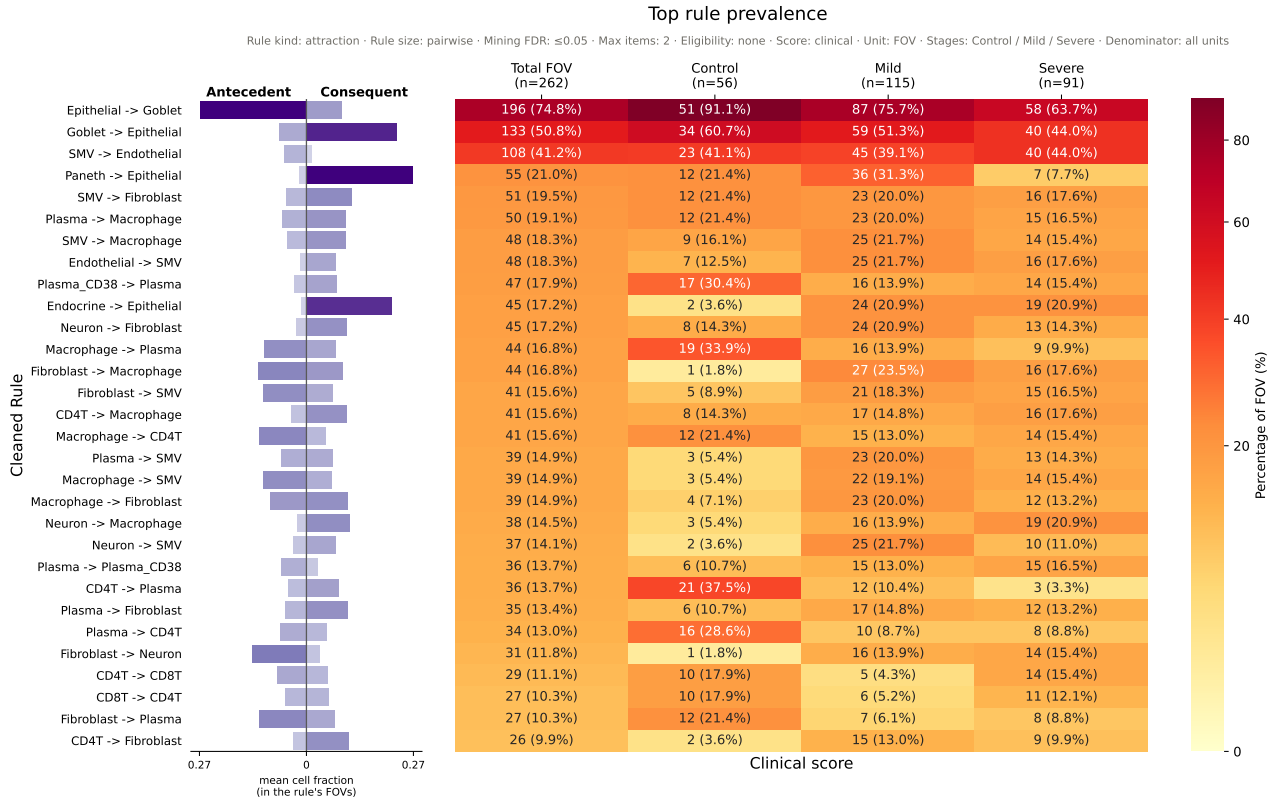


Figure 1: Rows are rules, ordered by how often they occur. Each cell gives the count and the percentage of units in that group holding the rule. The bars on the left are the mean share of cells of the antecedent (left) and consequent (right) type, over the FOVs where the rule fires. No eligibility filter: every unit is counted.

Counting each patient once instead of each FOV leaves the ordering and the stage pattern unchanged – Paneth → Epithelial 57% to 16% of patients, Endocrine → Epithelial 14% to 44%, CD4T → Plasma 57% to 8% – so none of this is an artefact of how many fields each patient contributed.

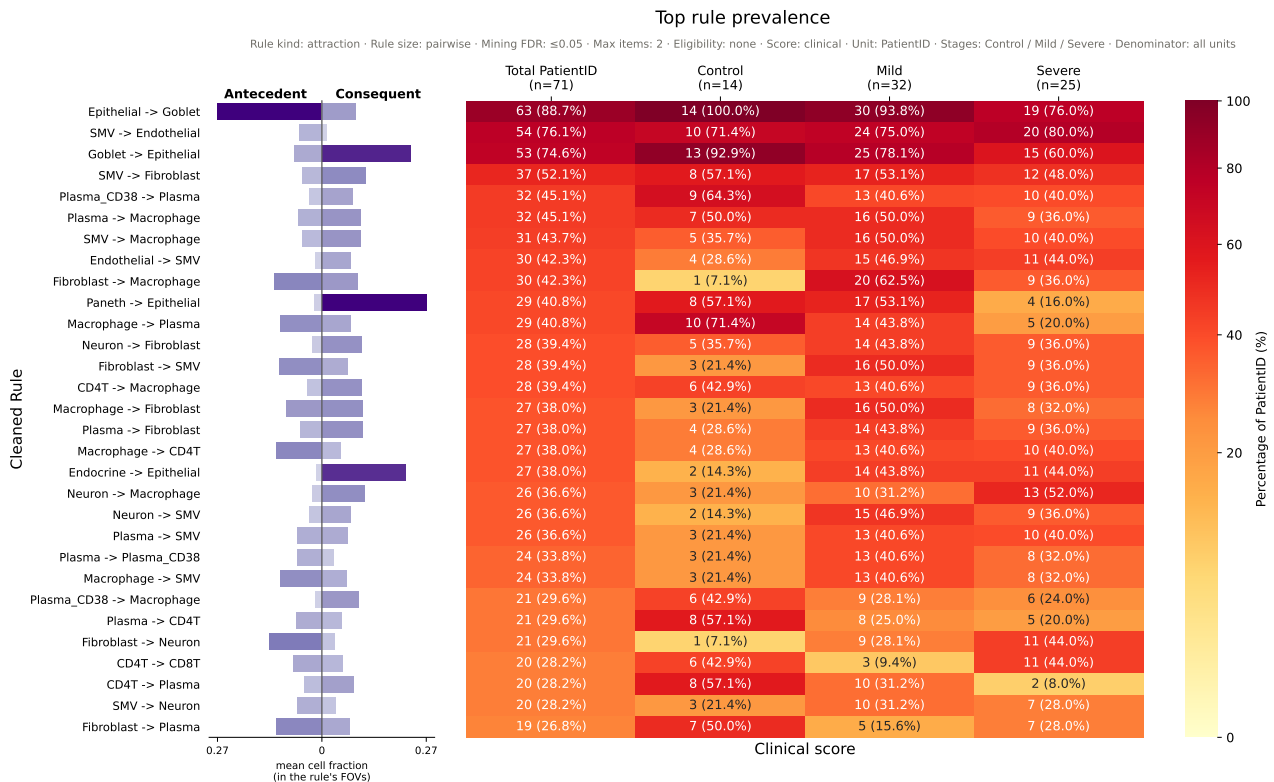


Figure 2: Values follow Figure 1; one patient is now one unit, so a rule counts once however many of that patient's fields hold it.

Organization moves much more than composition

This is the central control. The next two figures use the same FOVs, the same organ and the same two stages, measured first as rules and then as plain cell counts. The rules move by 37 to 67 percentage points; the cell counts behind them move by at most 9.

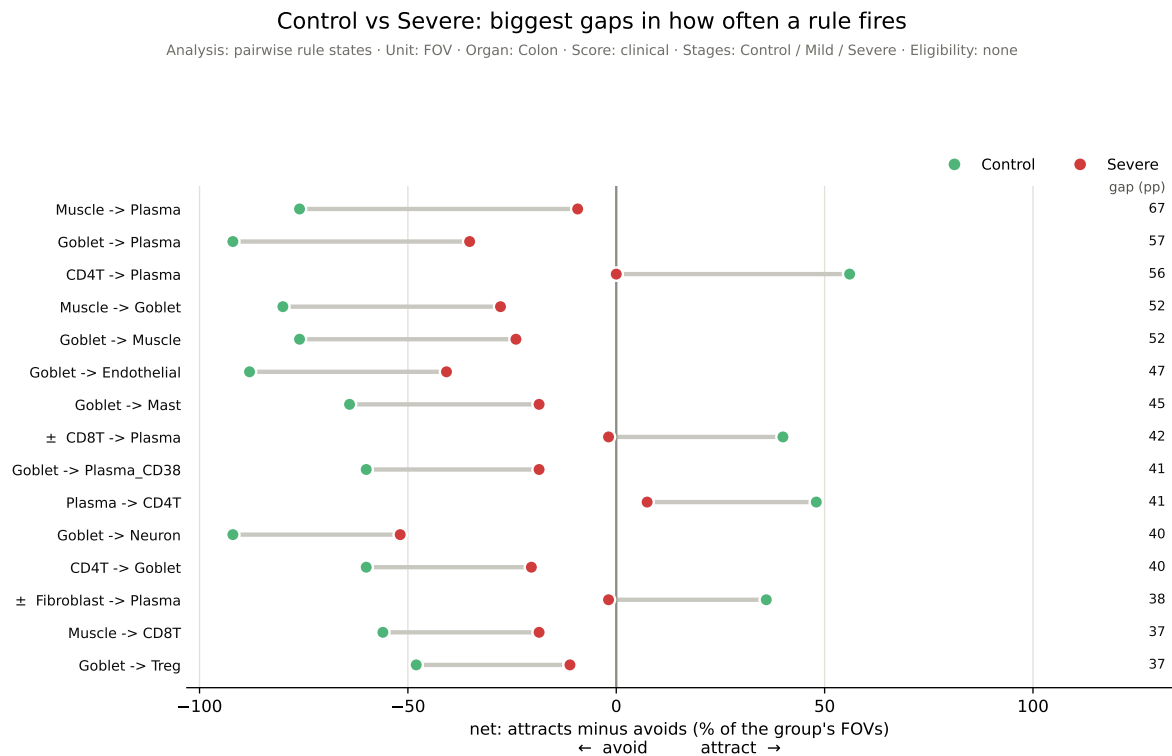


Figure 3: Each row is one rule; the two dots are its net score in the two groups and the bar is the gap in percentage points. Hollow dots and grey text mark rules that did not pass FDR 0.05. Rows are ordered by gap size. Colon, Control against clinically Severe, no eligibility filter. See the [test guide above](#).

Measured as plain cell abundance instead, the same fields barely move.

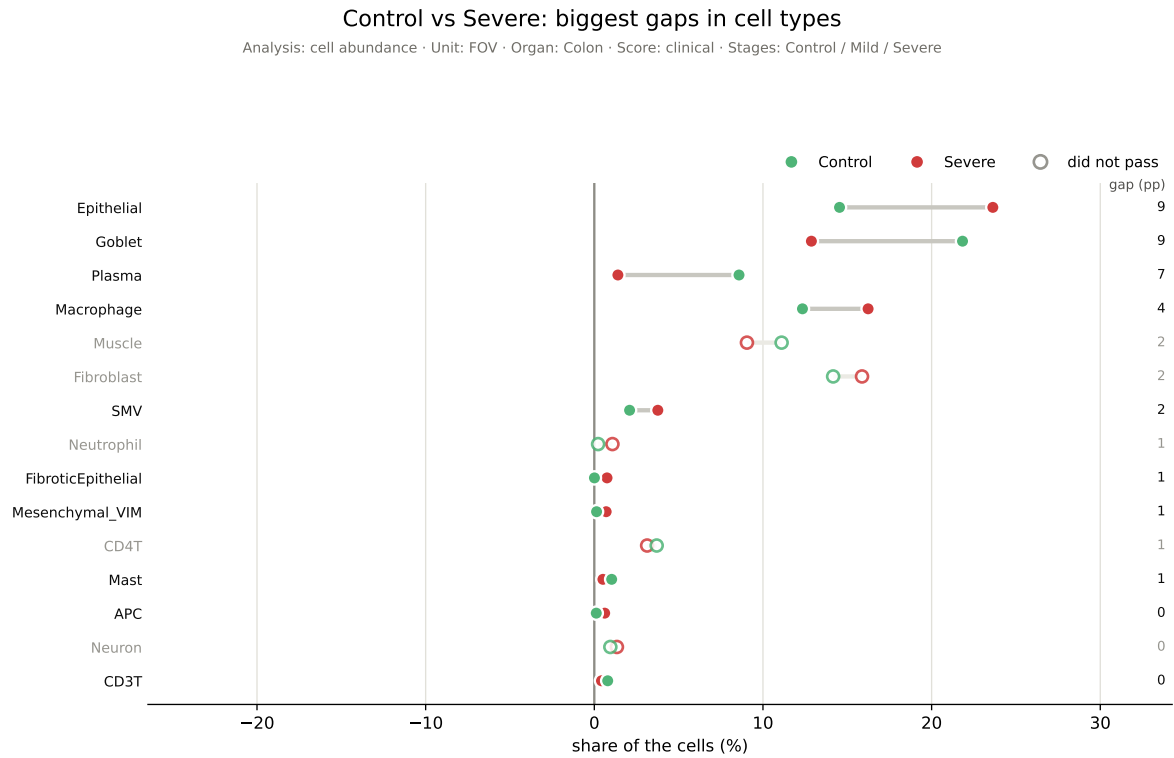


Figure 4: The same two groups of FOVs and the same layout as Figure 3, but each row is now a cell type and the value is its share of all cells in the FOV rather than a rule's net score.

The same contrast holds across all three stages at once. Every colonic rule that climbs is a goblet or muscle avoidance relaxing by 18 to 31 points per stage, while goblet cells fall 5 points and nothing else moves more than 3.

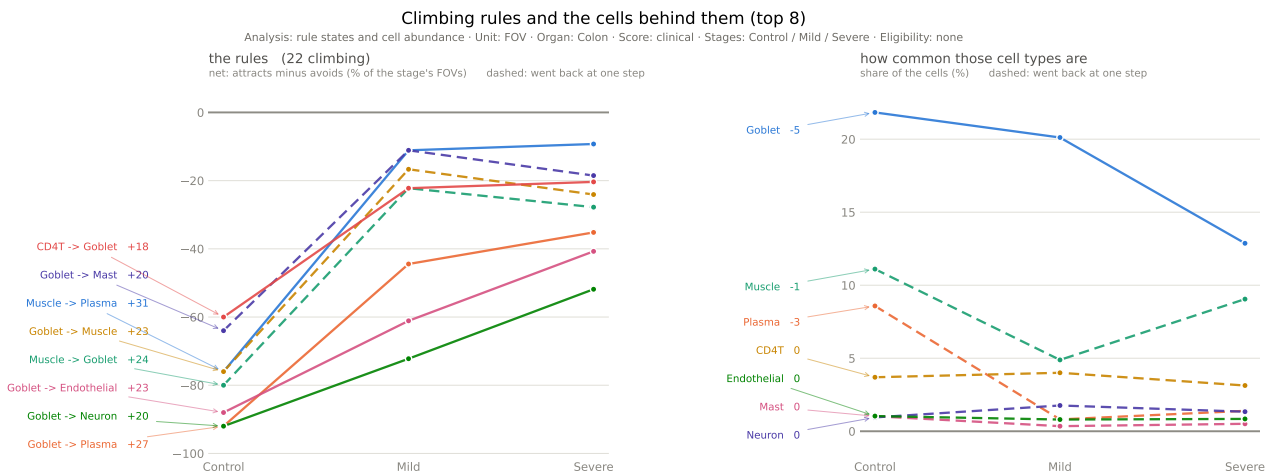


Figure 5: Left: the net score of each rule that moves steadily across the three stages, with its change per stage beside the name. Right: the mean share of the cell types those same rules name, with its own change per stage. Both panels are in percentage points, so the two can be compared directly. A dashed line moved back at one step instead of moving throughout. No eligibility filter. Colon, clinical score. Twenty-two rules climb and the eight largest are drawn.

The rules that fade tell a sharper story: all eleven end at plasma or CD4T, and all eleven collapse to zero by Mild and stay there. That is a step at the onset of disease, not a gradient with severity.

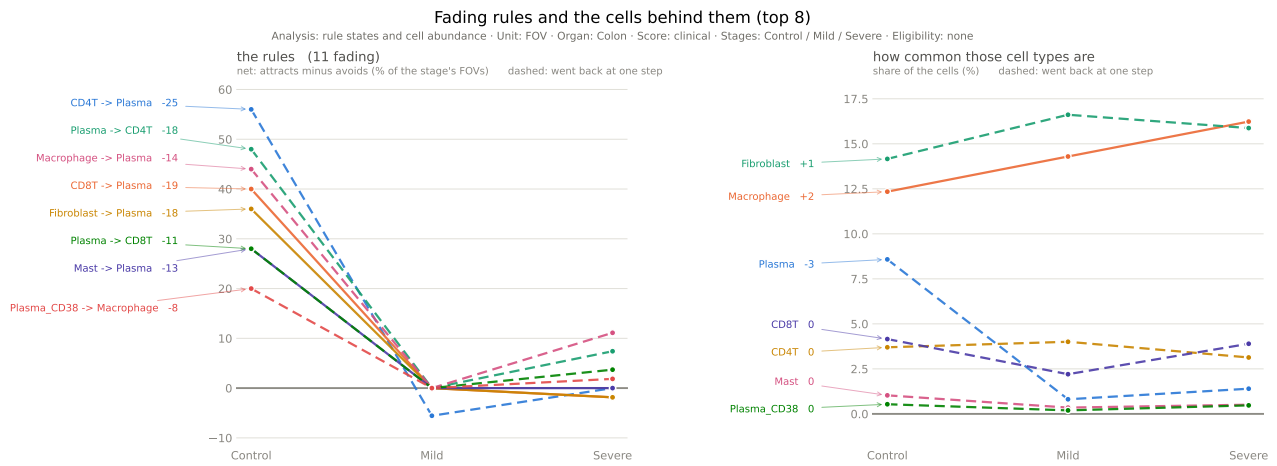


Figure 6: Values follow Figure 5, for the fading direction.

In the duodenum only two rules fade, and one of them moves against its own cells: fibroblast and muscle both grow, by 3 and 5 points, while Fibroblast → Muscle falls by 20 points per stage.

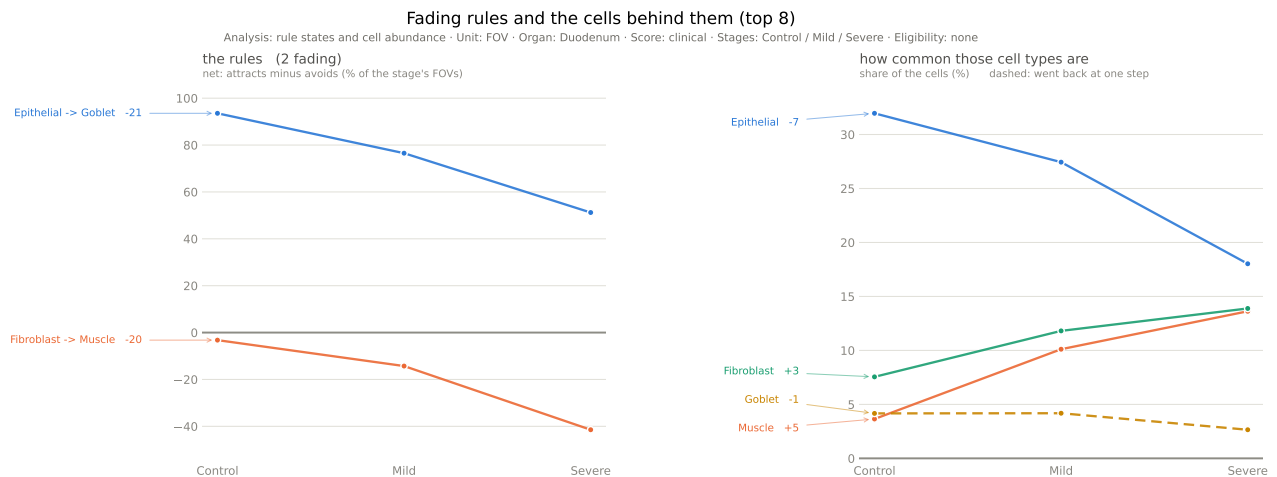


Figure 7: Values follow Figure 5; Duodenum.

Losing the cells is not the same as losing the arrangement

A rule can vanish for two very different reasons: the arrangement broke down, or the cells that made it are no longer there to count. Every rule therefore gets its own eligibility test, and the bottom panel keeps the unjudgeable fields visible as a separate grey block. Duodenal Epithelial → Goblet is the clean case: among fields that can still be judged the rule is as common in Severe disease as in Control (97%, 88%, 91%) and just as strong (Lift about 1.8 throughout), while the number of judgeable fields falls from 30 of 31 to 23 of 41.

Duodenum · Epithelial → Goblet

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.74 · C-M FDR 0.592 · C-S FDR 0.786 · M-S FDR 0.852

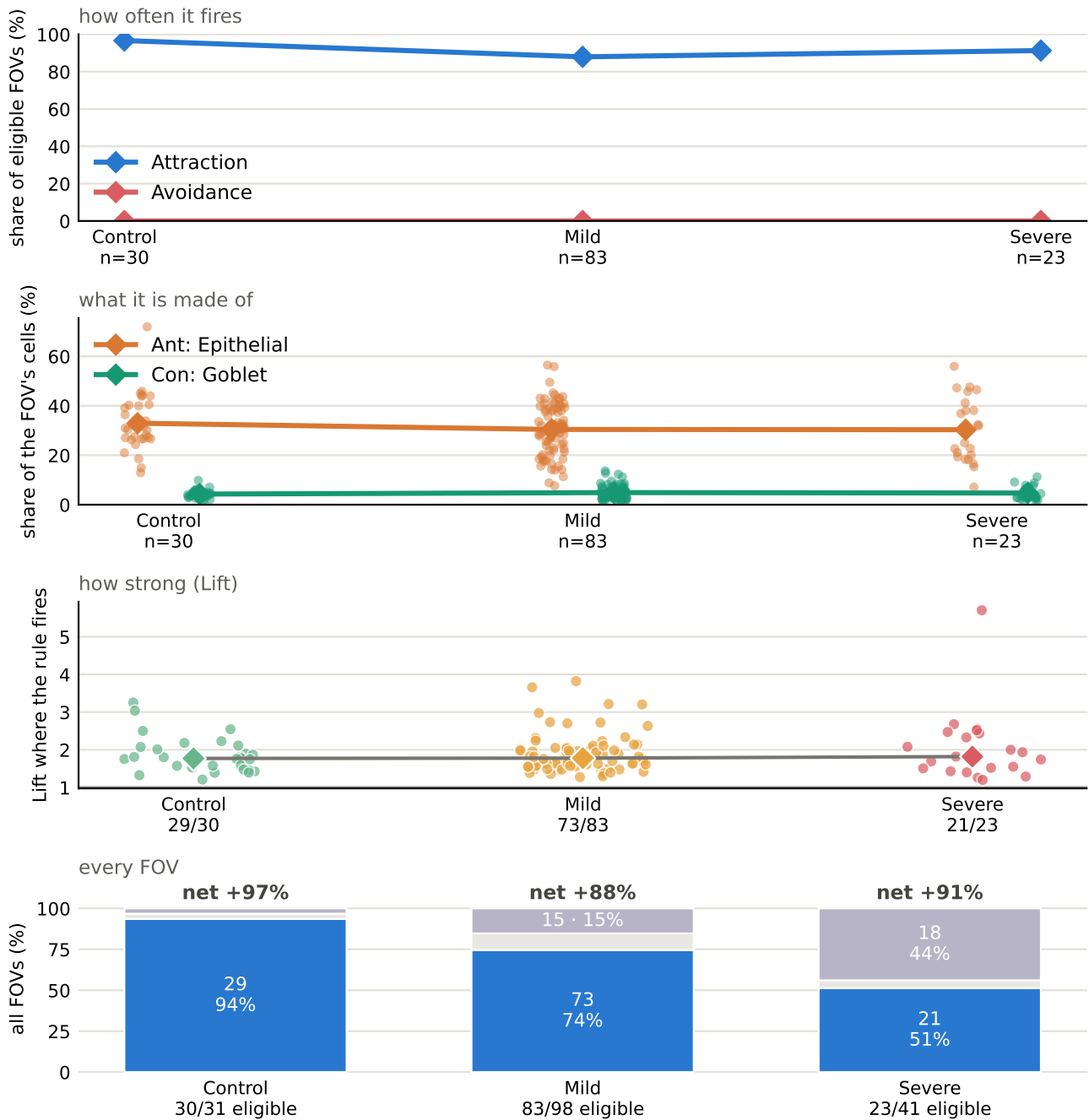


Figure 8: Four views of one rule, all on the same eligible FOVs. **How often it fires:** the share of that stage's eligible FOVs scoring attraction and the share scoring avoidance, as two separate lines, with the eligible count below. **What it is made of:** one dot per eligible FOV giving that FOV's share of cells belonging to each named type, with the stage mean as a diamond; the denominator is every cell in the FOV, not just the two types. **How strong:** one dot per rule-bearing FOV and the stage median, where Lift is observed co-occurrence divided by what independence predicts, so 1 is neutral, above 1 supports attraction and below 1 supports avoidance; the count below is rule-bearing / eligible. **Every FOV:** all FOVs of the stage split into attraction, no rule, avoidance and too few cells to judge, so composition loss stays visible instead of being scored as "no rule"; the net score sits above each bar. A field of view (FOV) is *eligible* for a rule only when it holds at least 20 cells of both named types. Among eligible FOVs the rule is attraction, avoidance, or no rule, and the net score is $100(N_{\text{attraction}} - N_{\text{avoidance}})/N_{\text{eligible}}$. See the [test guide](#) above.

That distinction does real work. Without the eligibility filter, 22 colonic rules climb and 11 fade with clinical severity (Figures 5 and following). With it, *none* of them passes – both panels below are empty. Colonic goblet cells fall from 22% to 13% of the tissue, so many fields stop being judgeable, and a rule going from avoidance to unjudgeable looks exactly like avoidance relaxing.

Rules that move with severity (top 15 each way)
 Analysis: pairwise rule states · Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type
 climbing with severity (0 rules) fading with severity (0 rules)

none

none

Figure 9: Each line is one rule's net score among eligible FOVs across the three stages, with its change per stage beside the name. Only rules passing FDR 0.05 in the ordered Control–Mild–Severe test are drawn, so an empty panel means nothing passed. Colon, clinical score, eligibility applied.

The duodenum keeps exactly two rules under the same test. Those two carry most of Part 3.

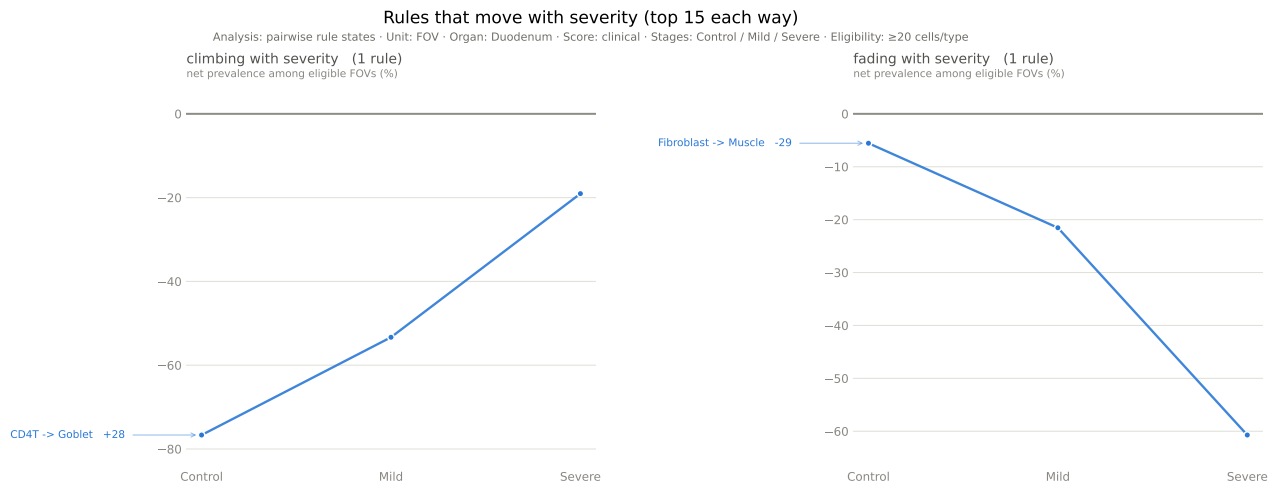


Figure 10: Values follow Figure 9; Duodenum.

The method recovers a boundary already known to be real

Colon and duodenum are different tissues, and the atlas treats them separately for that reason. A method that measures organization should rediscover the split without being told. It does, and all six differences that survive correction are anchored on goblet cells – the expected answer, since goblet cells are a fifth of all colonic cells and a twenty-fifth of duodenal ones.

From the atlas paper: “Because of the systematic changes between epithelial and immune composition and organization in the duodenum and the colon, we proceeded to analyze the duodenal and colonic aHSCt samples separately, comparing each with their relevant controls.”

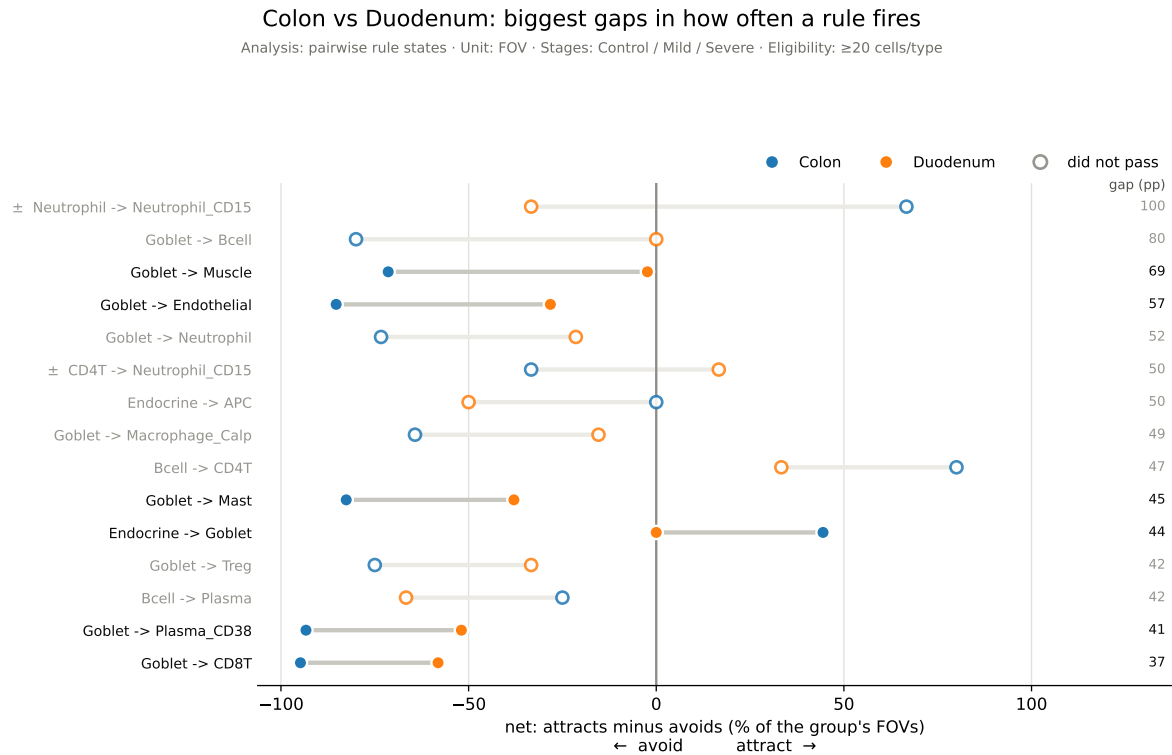


Figure 11: Values follow Figure 3; Colon against Duodenum with all stages pooled and the eligibility filter applied.

Rules are structured across patients

Summarising each patient by the share of their FOVs holding each rule, and then clustering, groups patients into recognisable profiles rather than scattering them: one block carries the stromal and macrophage rules, another the plasma and CD4T rules, and Epithelial $\rightarrow$ Goblet is close to universal.

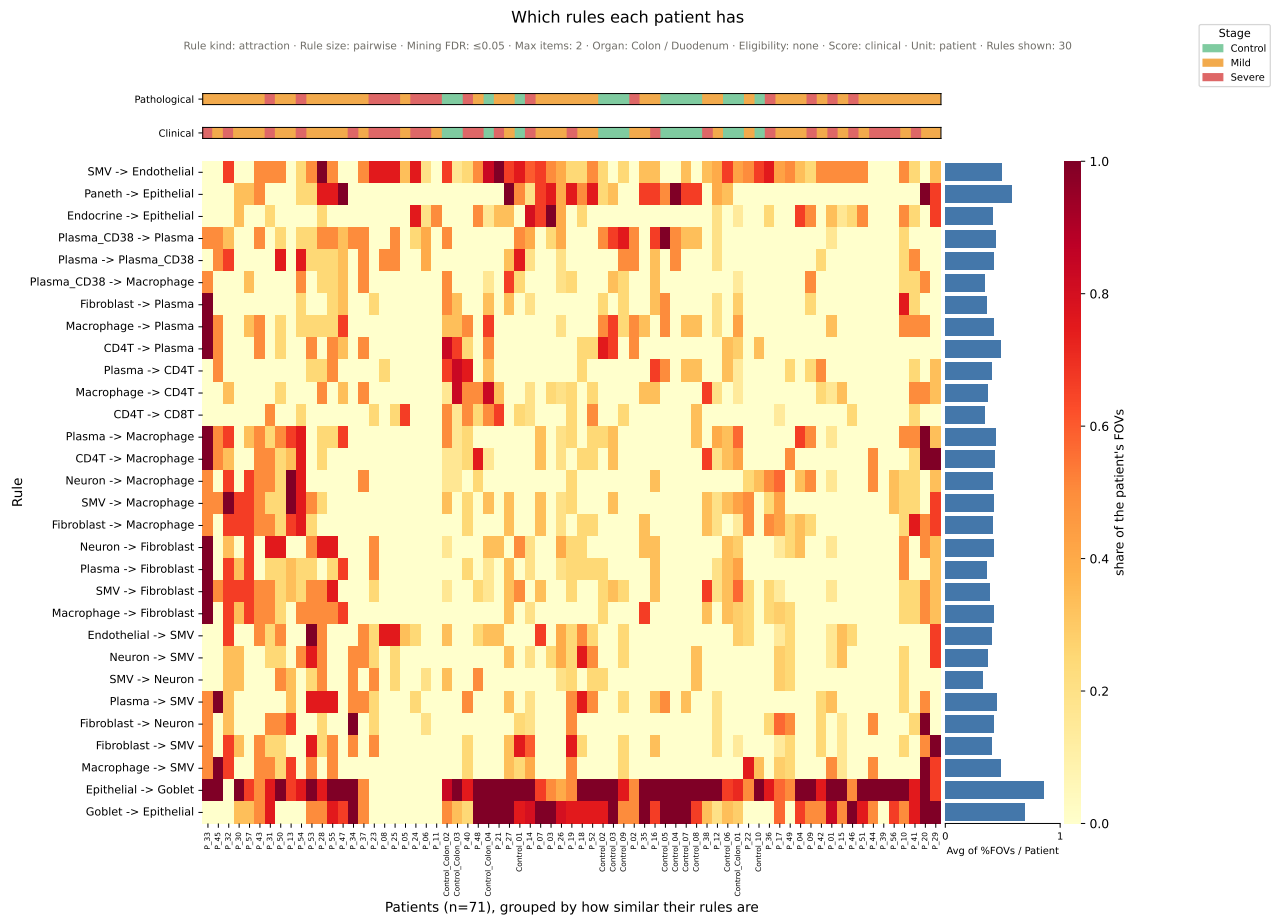


Figure 12: Rows are rules, columns are patients ordered by how similar their rule profiles are. Colour is the share of that patient's FOVs holding the rule. The two strips on top give each patient's pathological and clinical stage; the bars on the right are the rule's mean share across patients. No eligibility filter.

2 What agrees with the atlas

Where the atlas paper reports a finding, does the rule layer recover it independently? Each result below is paired with the sentence it corresponds to.

Paneth cells disappear from the duodenal crypt

Paneth cells sit at the base of the crypt and release antimicrobial peptides. Losing them is one of the few established tissue markers of gut aGVHD – the paper uses Paneth counts as the benchmark its own predictor has to beat. The count panel comes first because it decides what anything else can measure: the fields holding enough Paneth cells to judge fall from 13 of 31 to 13 of 41.

From the atlas paper: "In accordance with previous reports, we found a reduction in Paneth cells in a subset of patients but not in all."

Paneth cells available

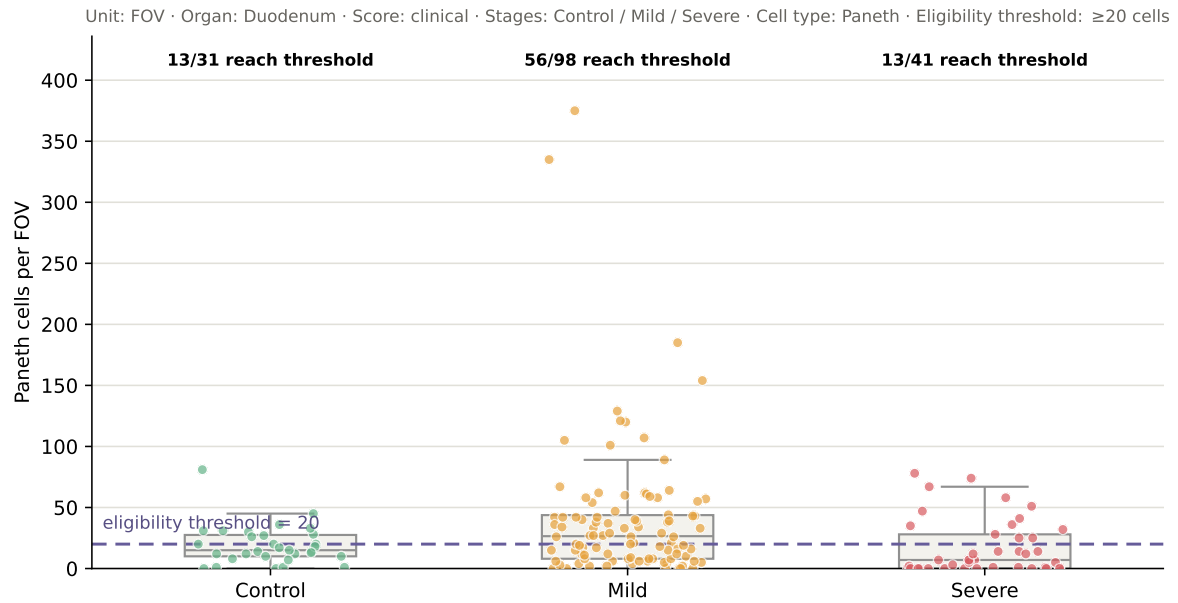


Figure 13: Every dot is one FOV's raw count of the named cell type, with the 20-cell eligibility threshold as a dashed line and the number of FOVs reaching it printed above each stage. This panel decides what the rule panels can measure at all. Duodenum, clinical stages.

Here the two severity scores part company, and the difference matters. Under pathological staging only 6 of 36 Severe fields remain judgeable and the rule's net score falls from +77% to +33%; under clinical staging 13 of 41 remain and the net barely moves. Paneth loss is a histological finding, and the pathological score is the one read off the same slide.

Duodenum · Paneth → Epithelial

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.16 · C-M FDR 0.447 · C-S FDR 0.395 · M-S FDR 0.557

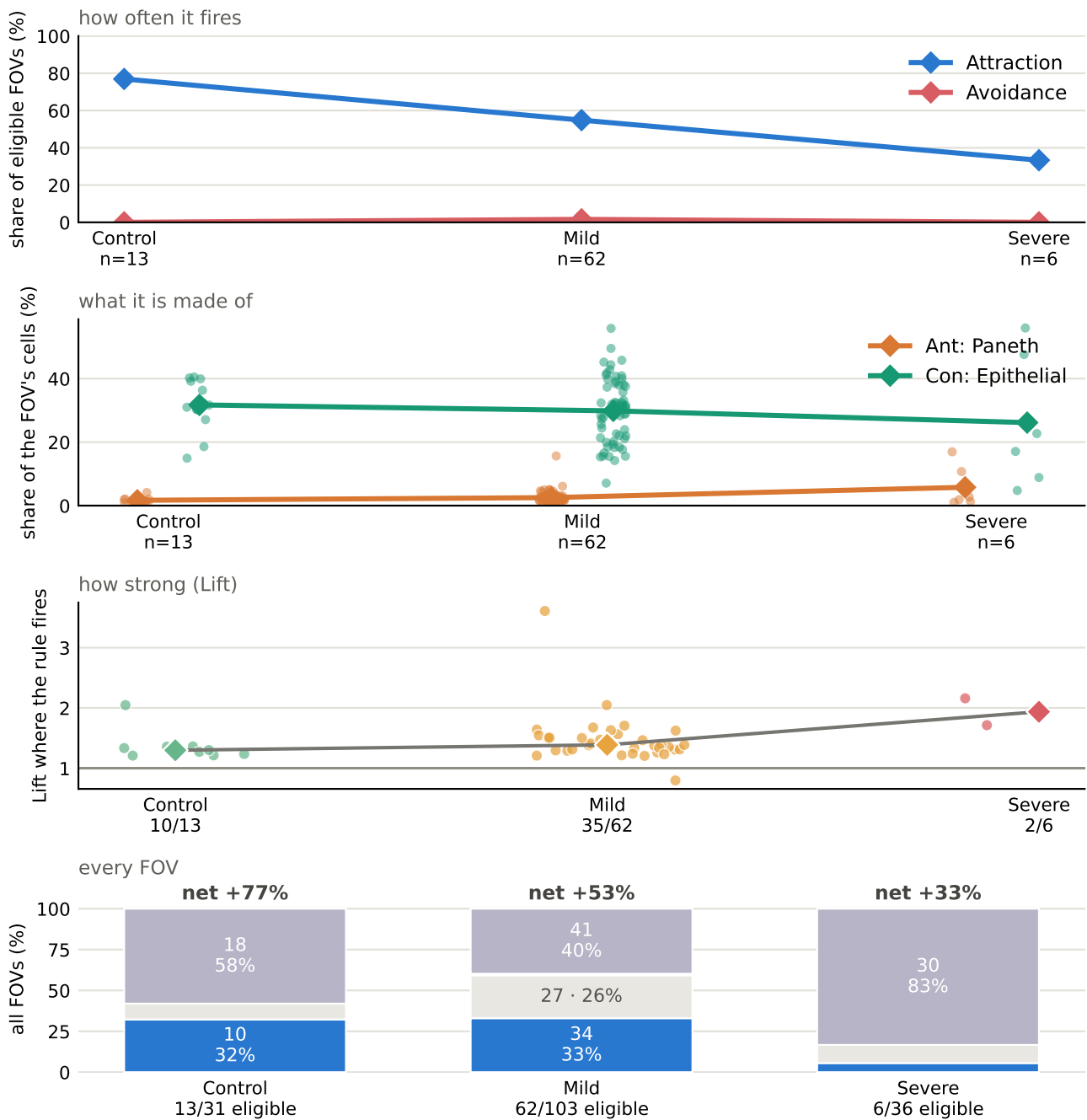


Figure 14: Values follow Figure 8; pathological stages.

Under clinical staging the collapse is much milder, and the rising Lift rests on 7 fields rather than 2.

Duodenum · Paneth → Epithelial

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.601 · C-M FDR 0.485 · C-S FDR 0.731 · M-S FDR 0.948

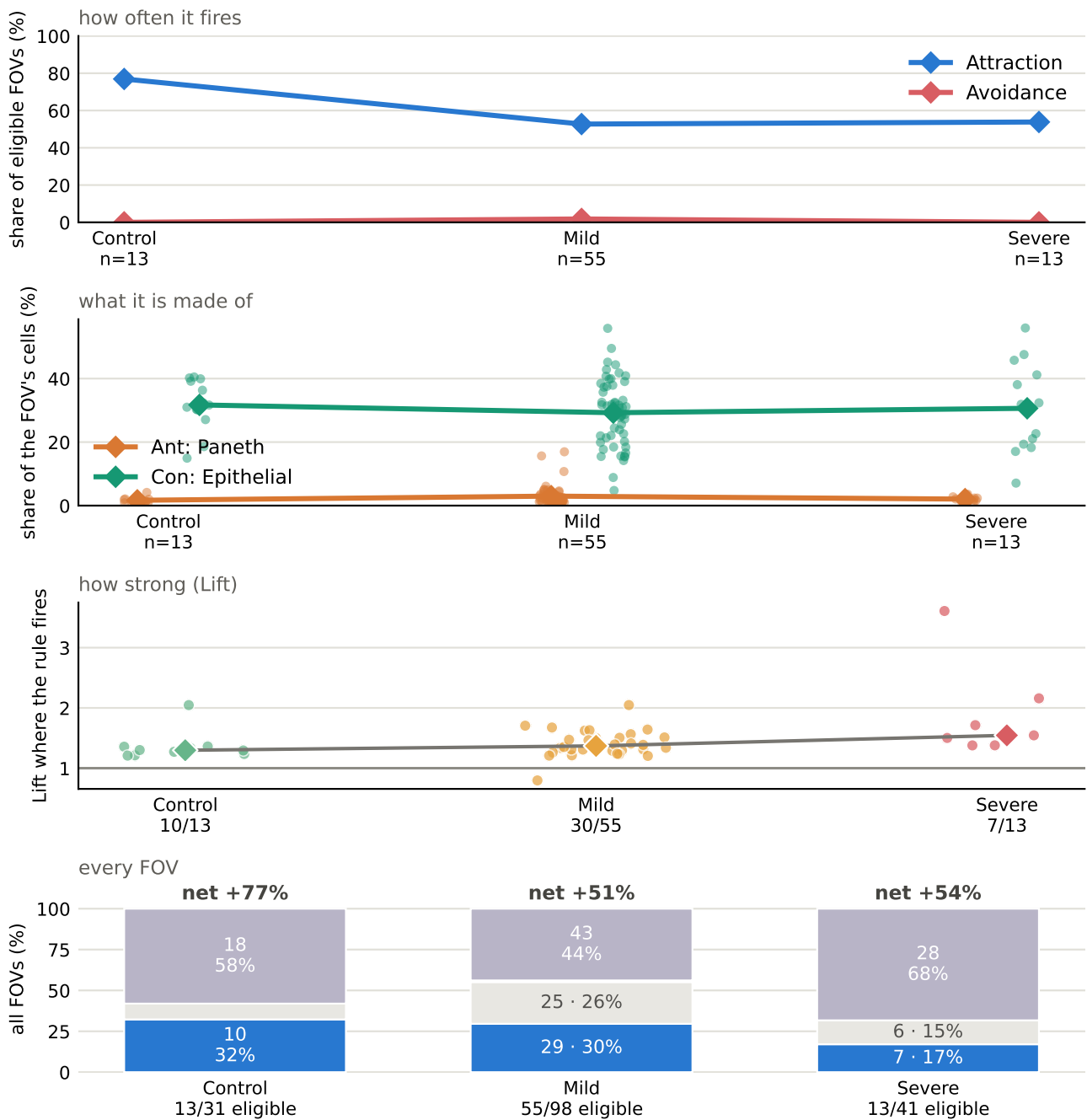


Figure 15: Values follow Figure 8; the same rule under clinical stages.

What is interesting is what happens where Paneth cells survive: their association with surrounding enterocytes does not weaken, it tightens. The reverse rule puts a corrected test behind that – Epithelial → Paneth climbs from a net of 0% to +50%, ordered FDR 0.029 – though its Severe endpoint is 3 fields out of 6.

Duodenum · Epithelial → Paneth

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.029 · C-M FDR 0.84 · C-S FDR 0.0561 · M-S FDR 0.0132

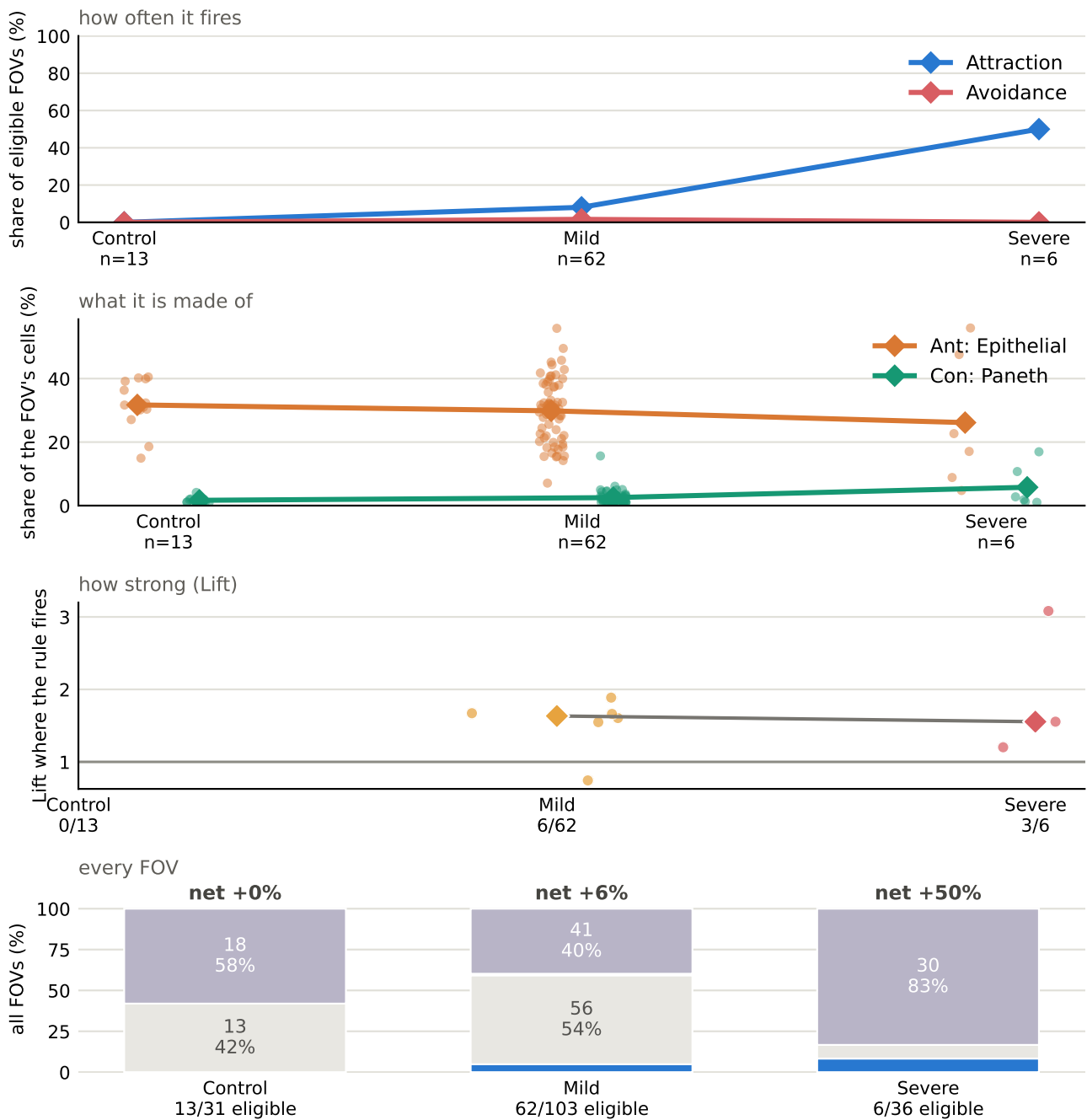


Figure 16: Values follow Figure 8; the reverse rule is counted and tested independently of the forward one, under pathological stages.

Endocrine cells accumulate where Paneth cells were

The paper's companion finding is that endocrine cells build up in the crypts, and it raises the possibility that stem cells misdifferentiate into endocrine cells instead of Paneth cells. The rule layer shows this as a threshold effect that is hard to argue with: not one control field in the whole duodenal cohort holds 20 endocrine cells, against 29 of 98 Mild and 8 of 41 Severe.

From the atlas paper: "Whereas in the normal duodenum, the endocrine cells were evenly spread along the villi, patients undergoing aHSCCT showed increased density primarily near the muscularis mucosa."

Endocrine cells available

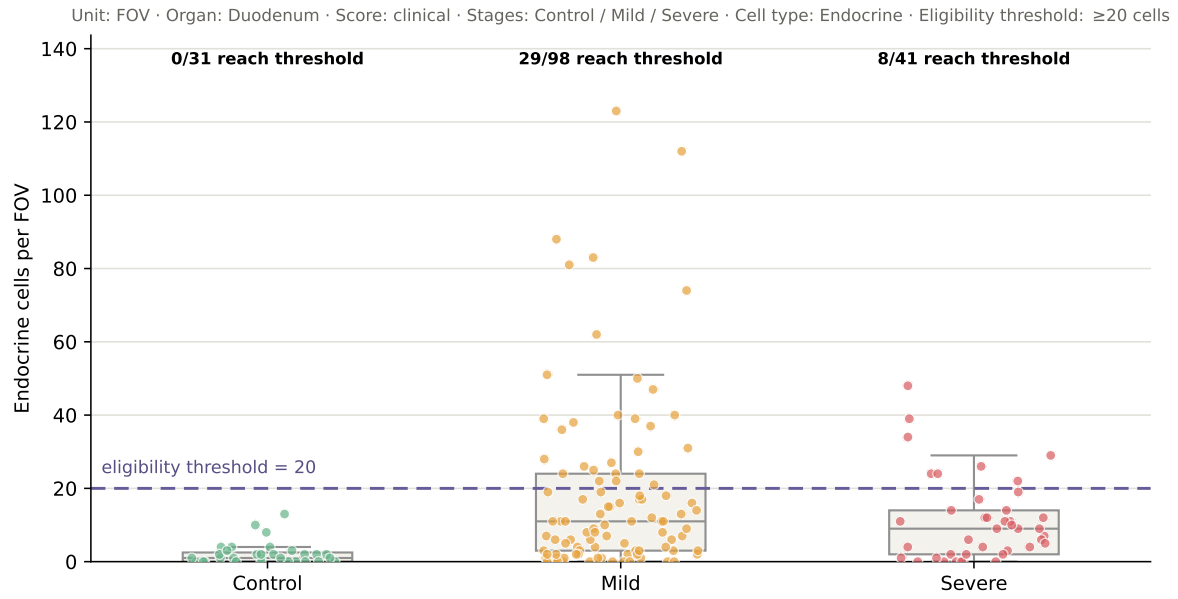


Figure 17: Values follow Figure 13, for endocrine cells.

Where the rule becomes measurable it is predominantly attractive, and both its prevalence and its strength rise with severity. This is the one finding here that is essentially unchanged between the two severity scores, because a threshold rather than a gradient drives it.

Duodenum · Endocrine → Epithelial

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: M-S FDR 0.42

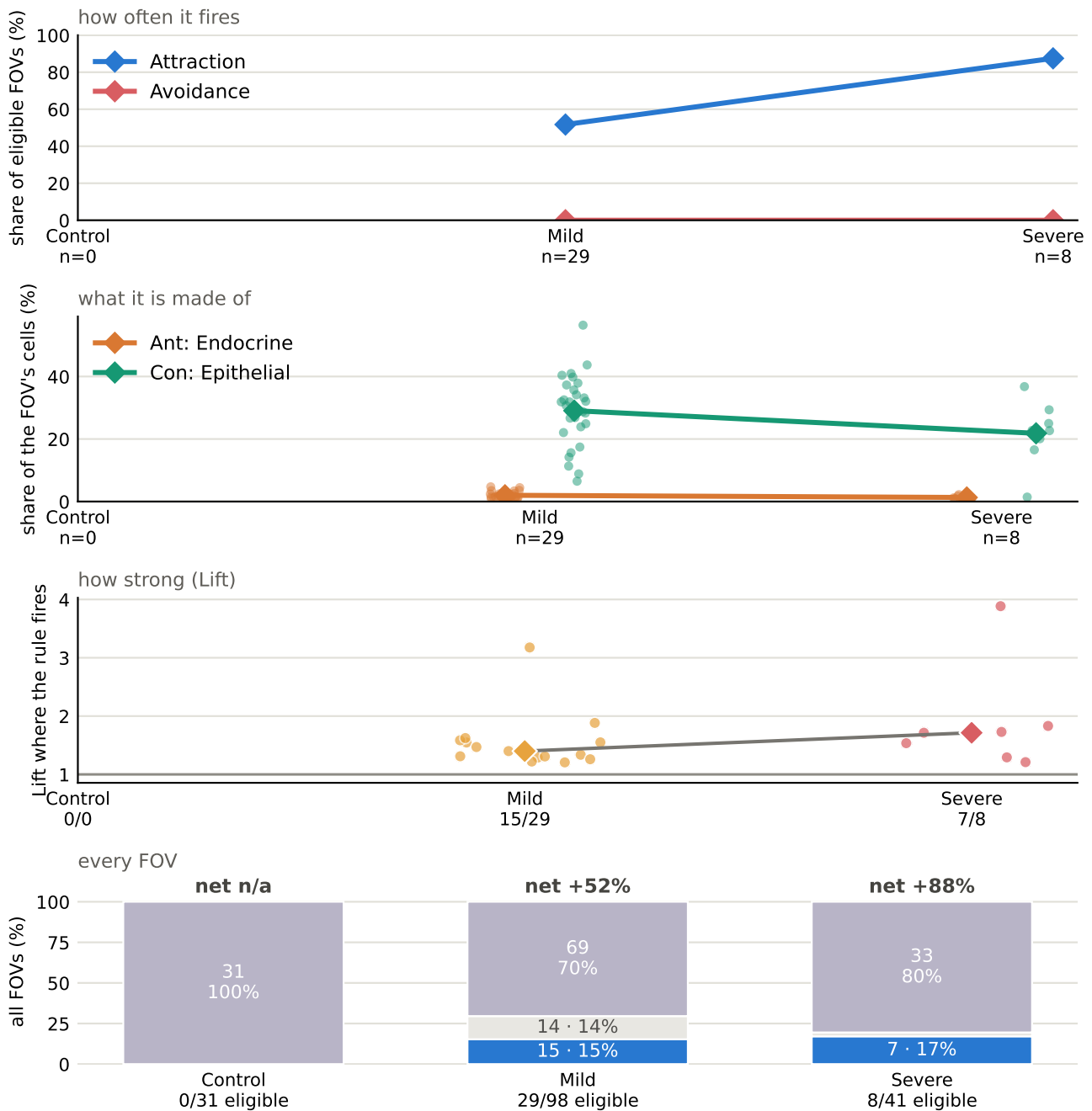


Figure 18: Values follow Figure 8. “n/a” in the bottom panel is not a missing value: with no eligible control field the rule cannot be evaluated there, so no Control-to-disease test exists.

Plasma cells lose their place in the colonic immune network

The loss of IgA-secreting plasma cells is the paper’s most emphatic immune finding, and it matters because those cells maintain the antibody barrier at the gut surface. The rule layer adds something the counts cannot: attraction falls from 56% of eligible fields to 0% while the plasma share of the tissue falls only from 8.5% to 3%, and it happens at Mild rather than progressively (C–S FDR 0.029).

From the atlas paper: “We found that patients undergoing aHSCT were characterized by a marked decrease in plasma and CD4 T cells and an increase in various stromal cells, mainly fibroblasts, in agreement with previous observations.”

Colon · CD4T → Plasma

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.116 · C-M FDR 0.0896 · C-S FDR 0.029

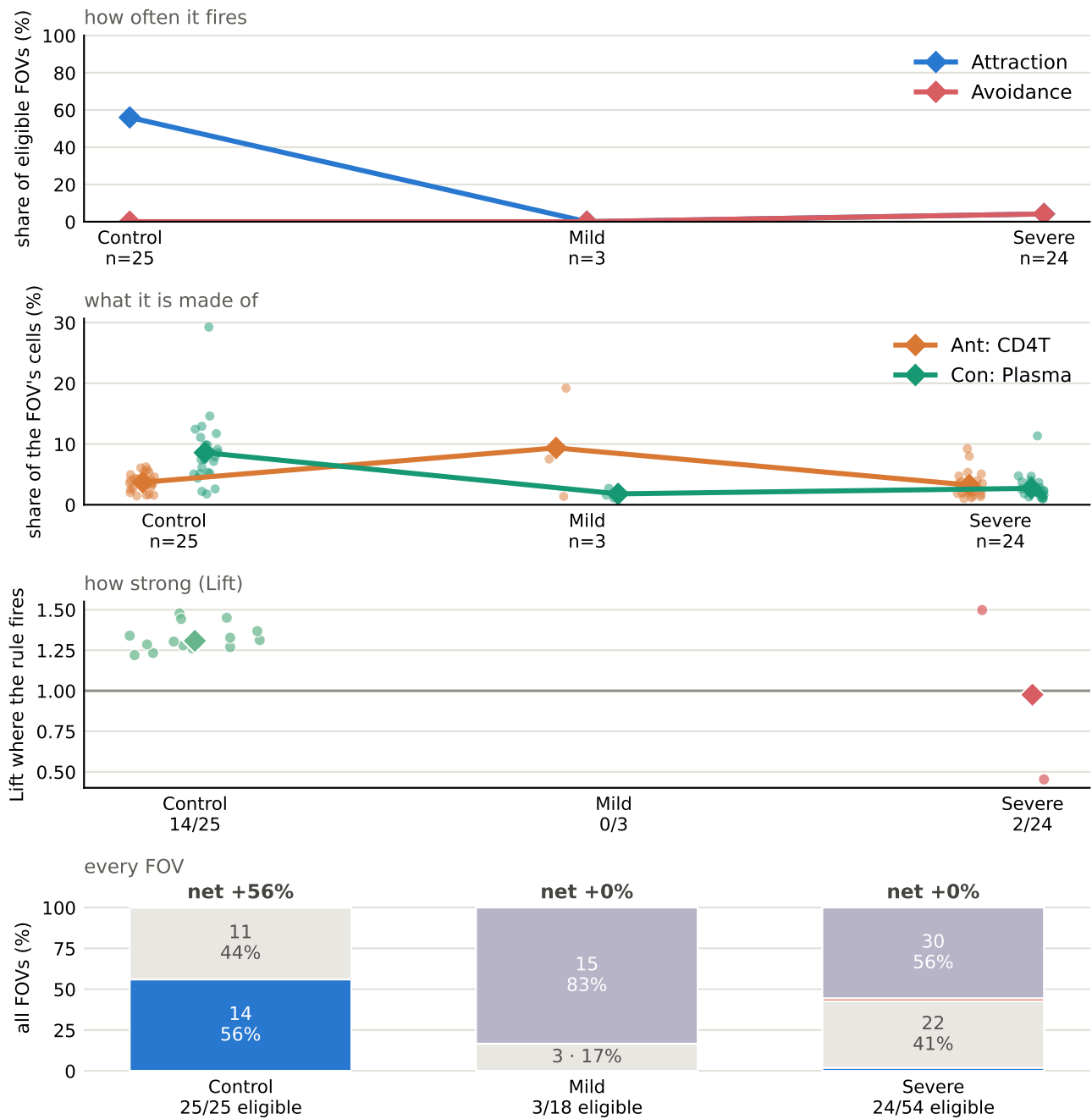


Figure 19: Values follow Figure 8. The Mild group holds 3 eligible fields, which the top two panels state under the axis.

CD8 T cells do the same thing at the same point, 40% to 0%, also at C-S FDR 0.029.

Colon · CD8T → Plasma

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: C-S FDR 0.029

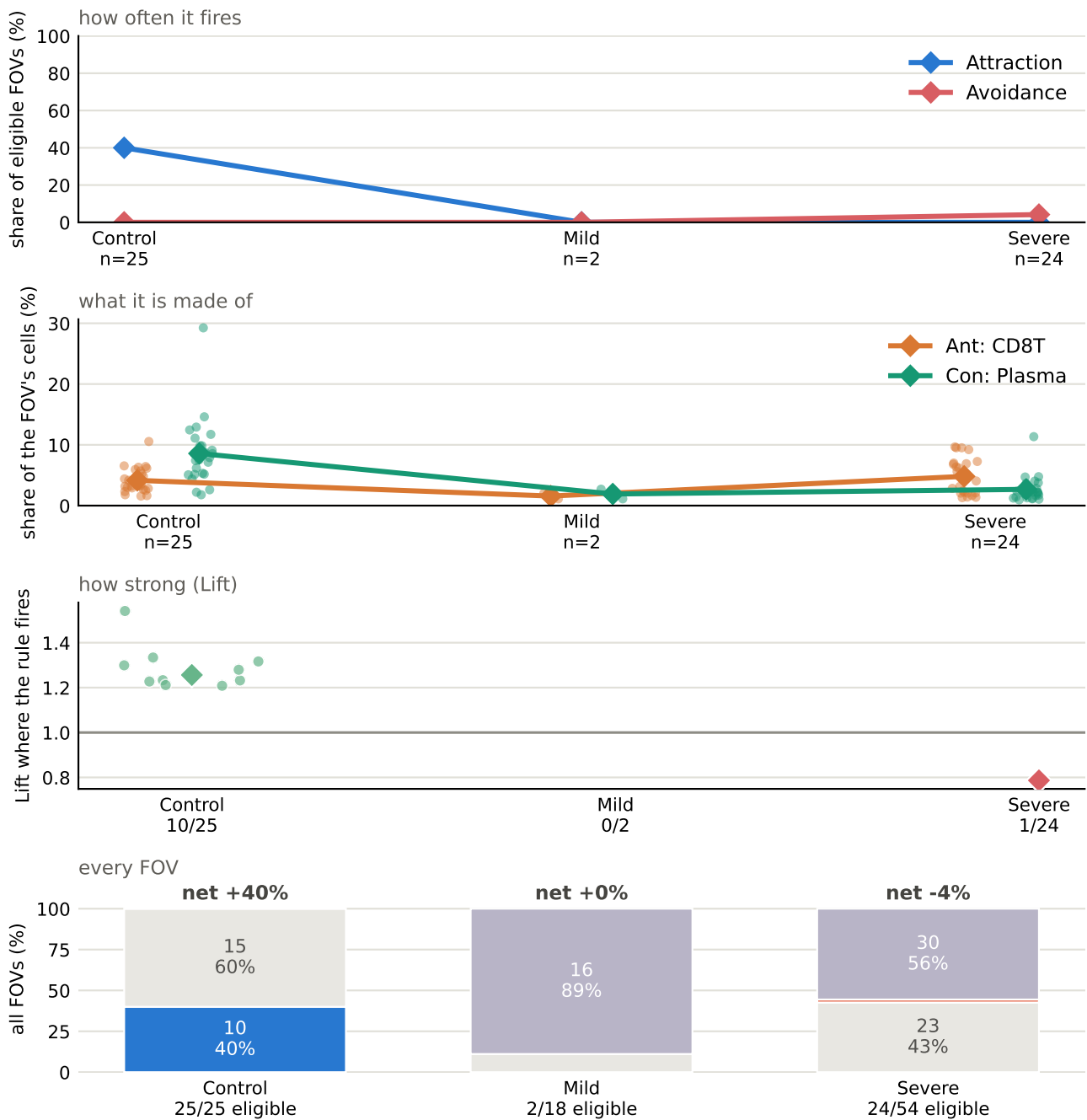


Figure 20: Values follow Figure 8.

This is where the clinical score earns its place as the default: both of these pass under clinical staging and neither is selected under pathological. The paper reaches its immune conclusions from the clinical score too.

Stroma takes over as the epithelium retreats

The paper reports fibrosis and rising fibroblasts alongside epithelial loss. The rule table shows the stromal side appearing where it was absent: Fibroblast → Macrophage is found in 1 of 56 control fields and in 27 of 115 Mild and 16 of 91 Severe ones (Figure 1). The corrected duodenal test picks up the matching rearrangement, and this one holds under both scores: avoidance rises from 6% to 61% of eligible fields (ordered FDR 0.022) while fibroblast and muscle shares both grow, and while the unjudgeable block *shrinks* from 42% to 32% – so it cannot be an eligibility artefact.

From the atlas paper: “a quantification of cell types across patients undergoing aHST showed a high degree of variability in epithelial cells, with many patients exhibiting a major decrease, which, in the duodenum, was also accompanied by an increase in fibroblasts and extracellular matrix (ECM) and correlated with high-stage disease.”

Duodenum · Fibroblast → Muscle

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.0217 · C-M FDR 0.609 · C-S FDR 0.0557 · M-S FDR 0.0557

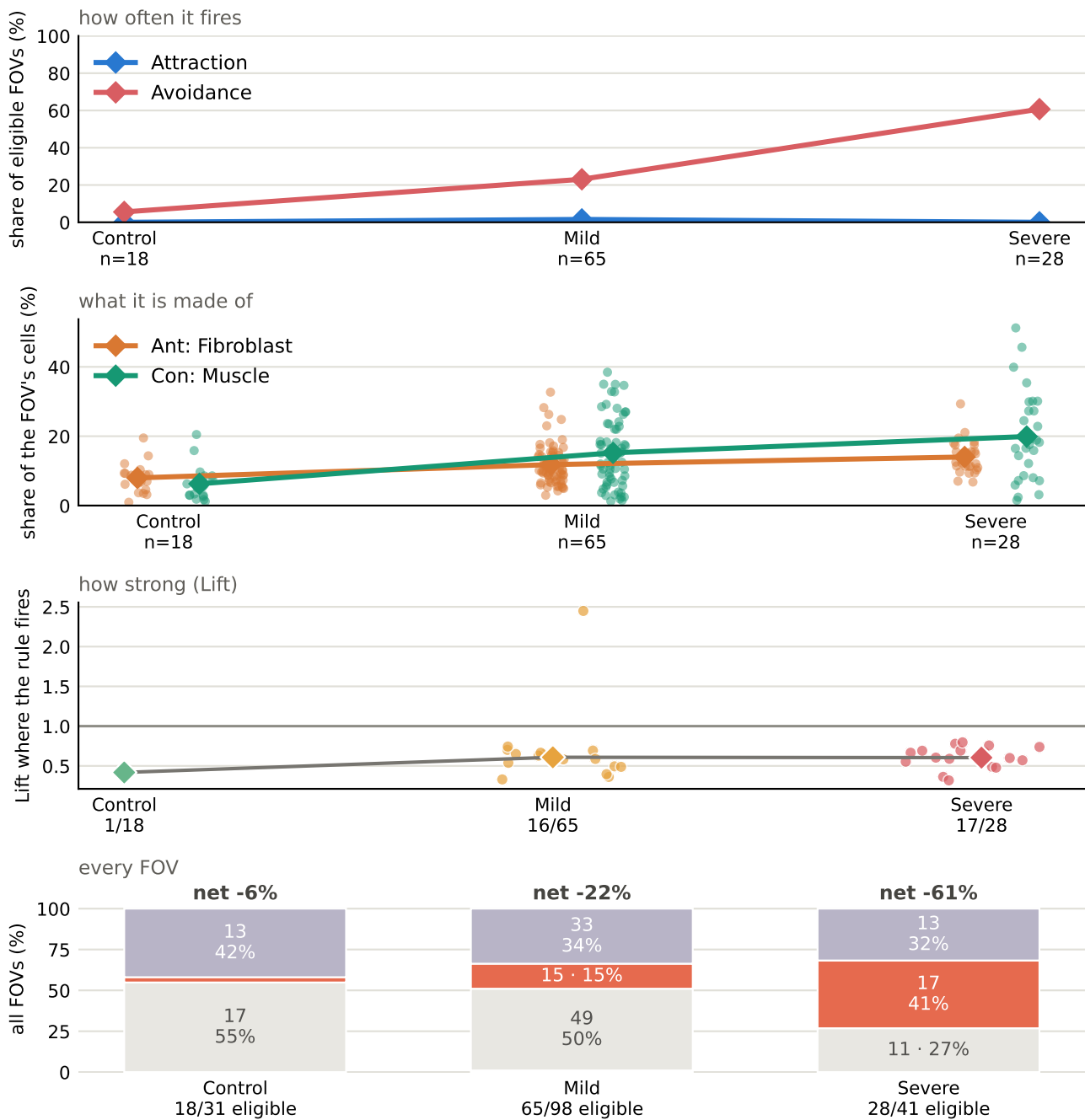


Figure 21: Values follow Figure 8.

In tissue this is a dense muscle mass with fibroblasts occupying the space around it; the two barely mix, and the separation deepens as the muscle mass grows.

Duodenum Fibroblast → Muscle — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength rule-bearing FOV in each stage

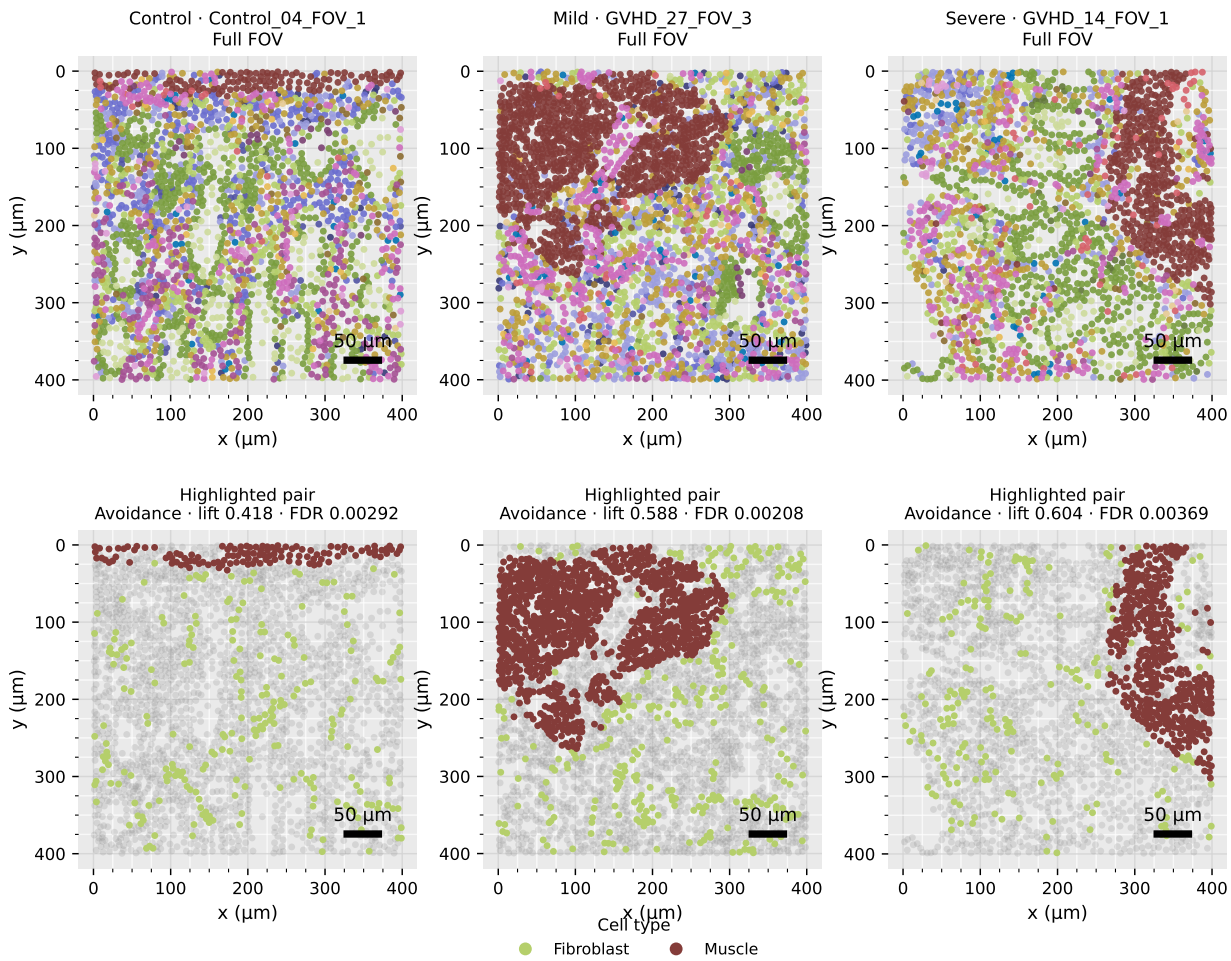


Figure 22: The top row shows every cell in the FOV; the bottom row greys out all cells except the two named types. Within each stage the FOV is picked automatically: take the more frequent rule direction among eligible rule-bearing FOVs, then the occurrence whose Lift is closest to that stage's median. The state, Lift and mining FDR above the bottom map belong to that one FOV, not to the between-stage test. These maps illustrate; the aggregate figure before them is the evidence.

Colonic goblet cells, and why the score matters

In healthy colon goblet cells are packed with mucus and hold everything else out; the screen sees near-total avoidance, with net scores close to -100% . The paper reports that this population is damaged in colonic aGVHD, and if goblet cells lose their mucus the exclusion should weaken. It does – but how it weakens depends entirely on which score is used, and that is worth stating plainly rather than choosing the friendlier version.

From the atlas paper: “In a subset of colon samples, we observed a loss of mucin expression in the epithelium, indicating potential damage to goblet cells or impaired mucin production.”

Under pathological staging the relaxation is a clean gradient: avoidance 95%, 72%, 27% of eligible fields, ordered FDR 0.016. Note the grey unjudgeable block growing from 20% to 66% underneath it.

Colon · Goblet → Muscle

Unit: FOV · Organ: Colon · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.0155 · C-M FDR 0.281 · C-S FDR 0.0161 · M-S FDR 0.151

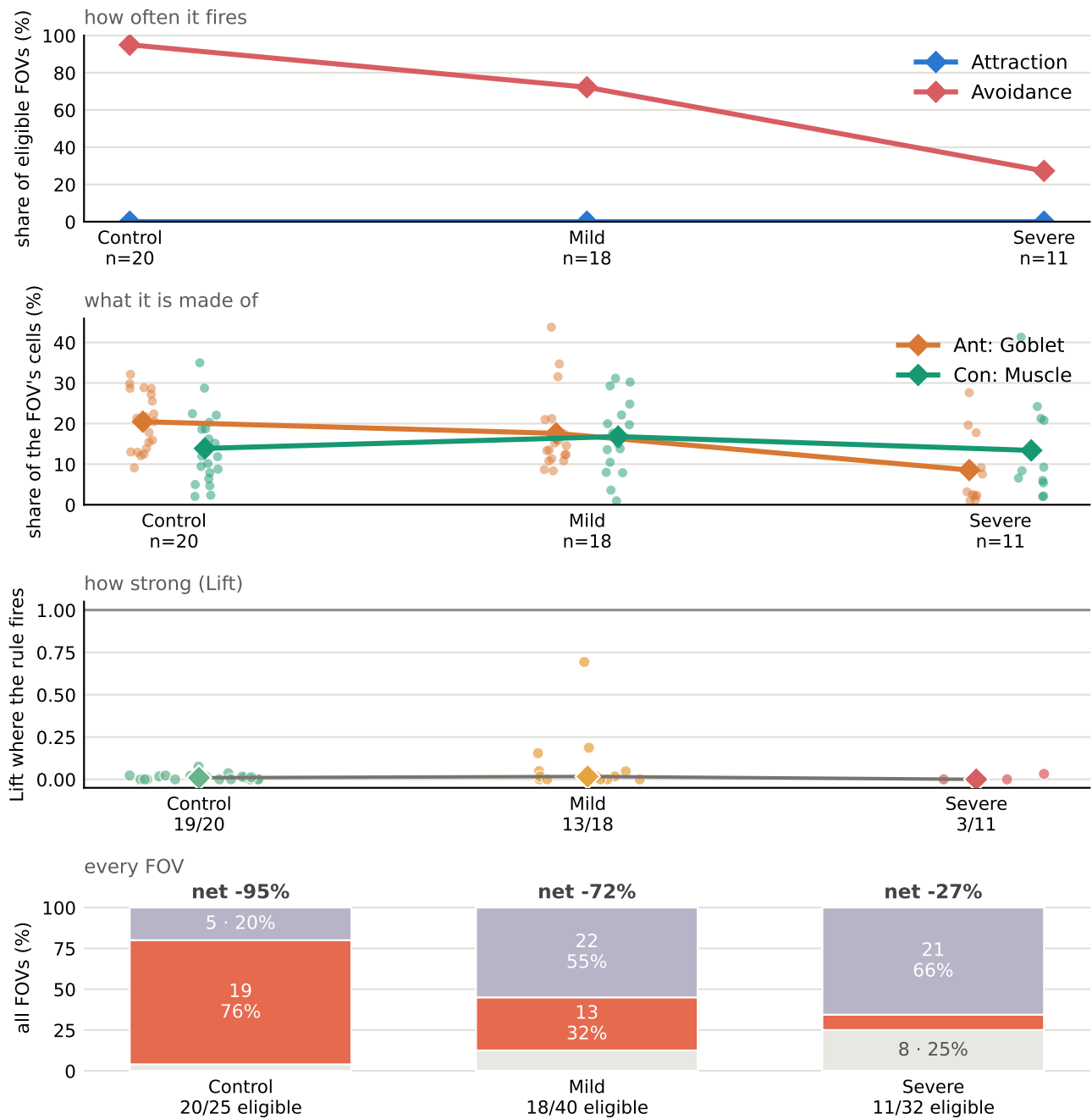


Figure 23: Values follow Figure 8; pathological stages.

Under clinical staging the same rule drops from 95% to 50% at Mild and then stops, at 57% in Severe. Ordered FDR 0.175, C-M FDR 0.041: a step at the onset of disease rather than a gradient with severity.

Colon · Goblet → Muscle

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.175 · C-M FDR 0.0414 · C-S FDR 0.128 · M-S FDR 0.963

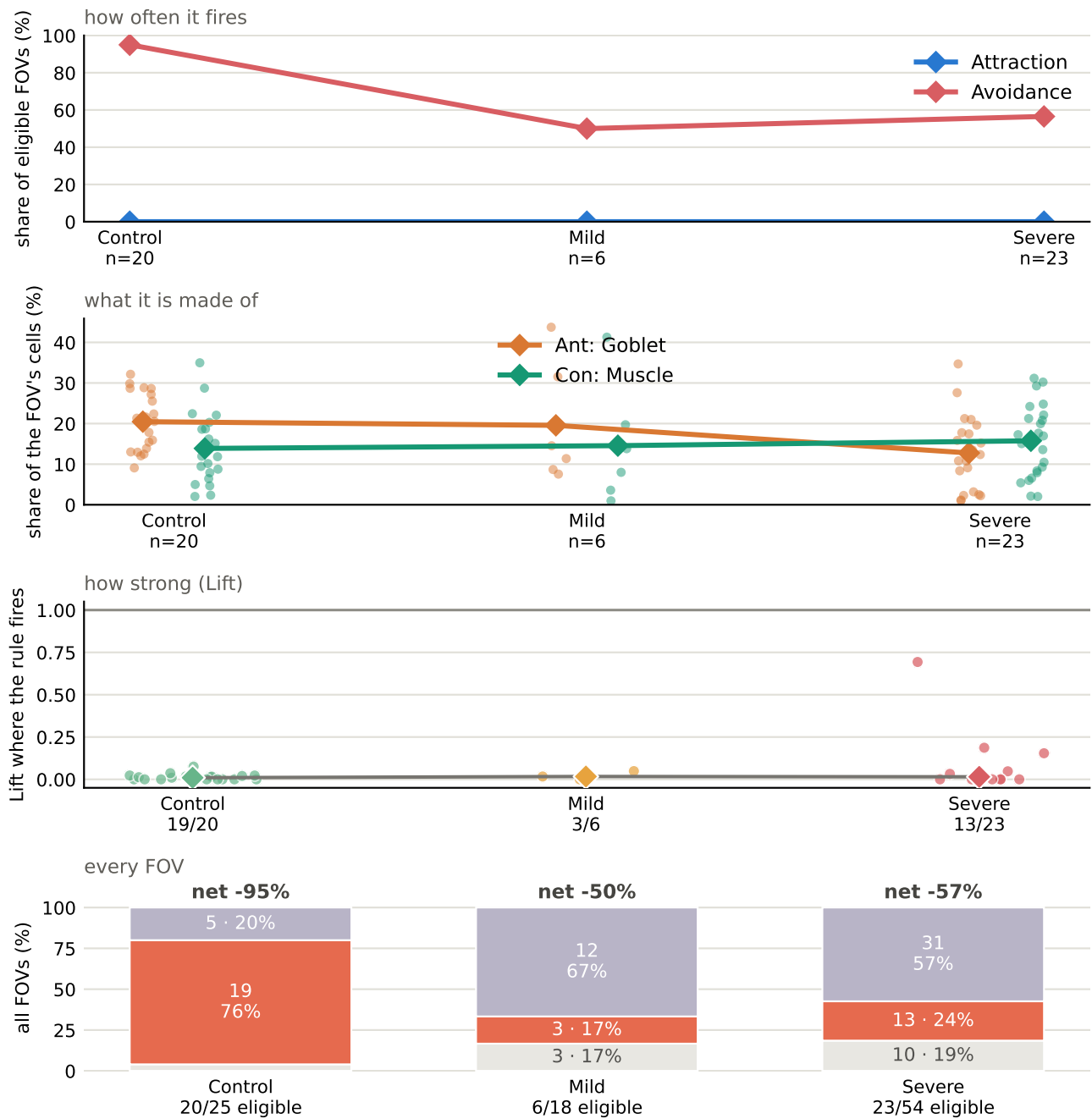


Figure 24: Values follow Figure 8; the same rule under clinical stages.

The reverse direction, calculated independently, moves the same way and by a similar amount: avoidance 100%, 67%, 65% of eligible fields, C-M FDR 0.029.

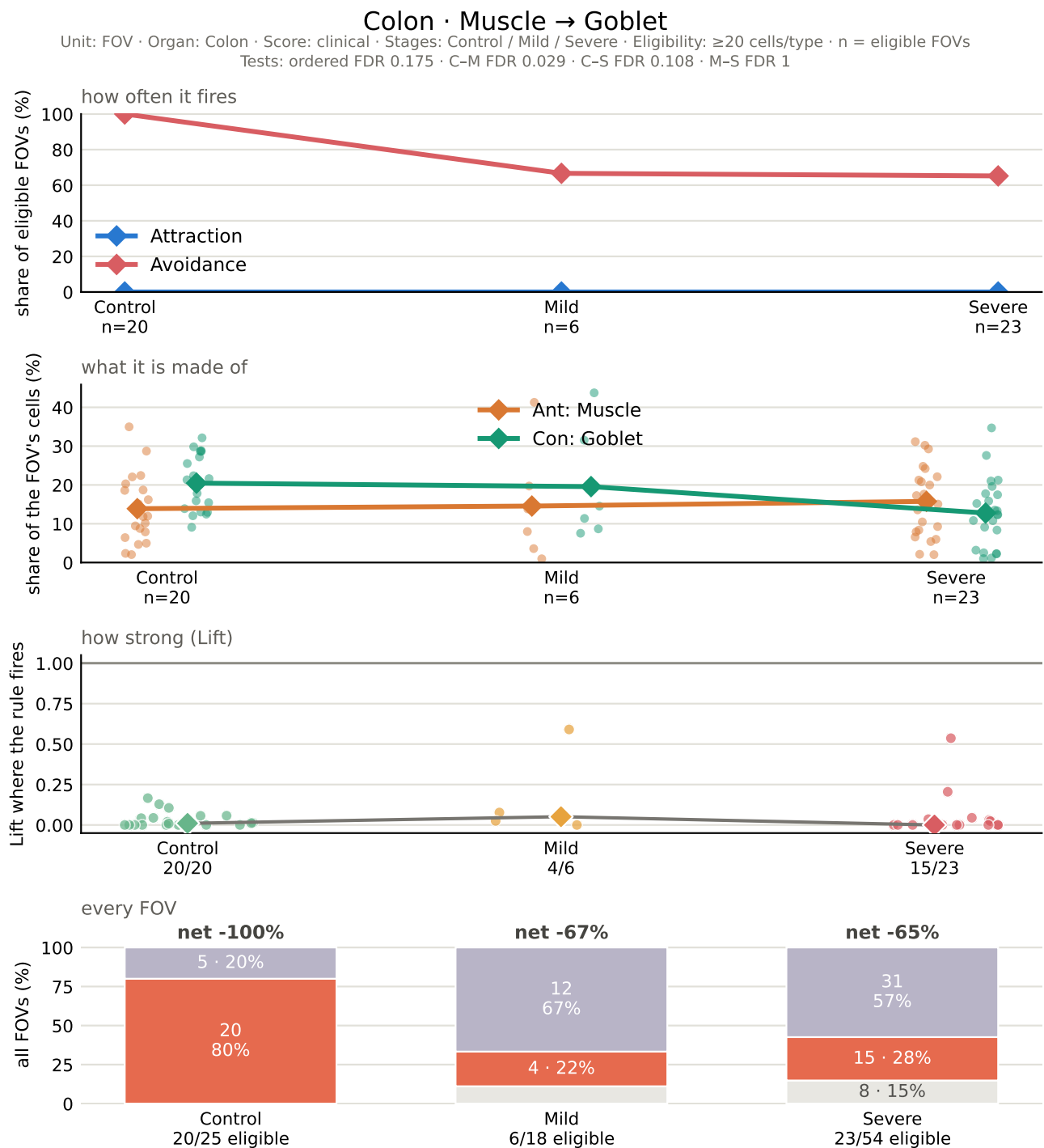


Figure 25: Values follow Figure 8; clinical stages.

CD8 T cells pull away from the epithelium

CD8 T cells are the classical effectors of aGVHD, and in healthy duodenum they are the population that sits *inside* the epithelium. The paper reports that their distribution around individual crypts becomes highly uneven in disease. Under pathological staging avoidance jumps to 75% of eligible fields in Severe (C-S FDR 0.013) while both populations stay present – but Severe is 8 eligible fields, of which 6 avoid.

From the atlas paper: “CD8 T and double-negative T cells (CD3 T cells and double negative for CD4 and CD8) were the predominant populations inside the epithelium.”

Duodenum · CD8T → Goblet

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.273 · C-M FDR 1 · C-S FDR 0.0132 · M-S FDR 0.0245

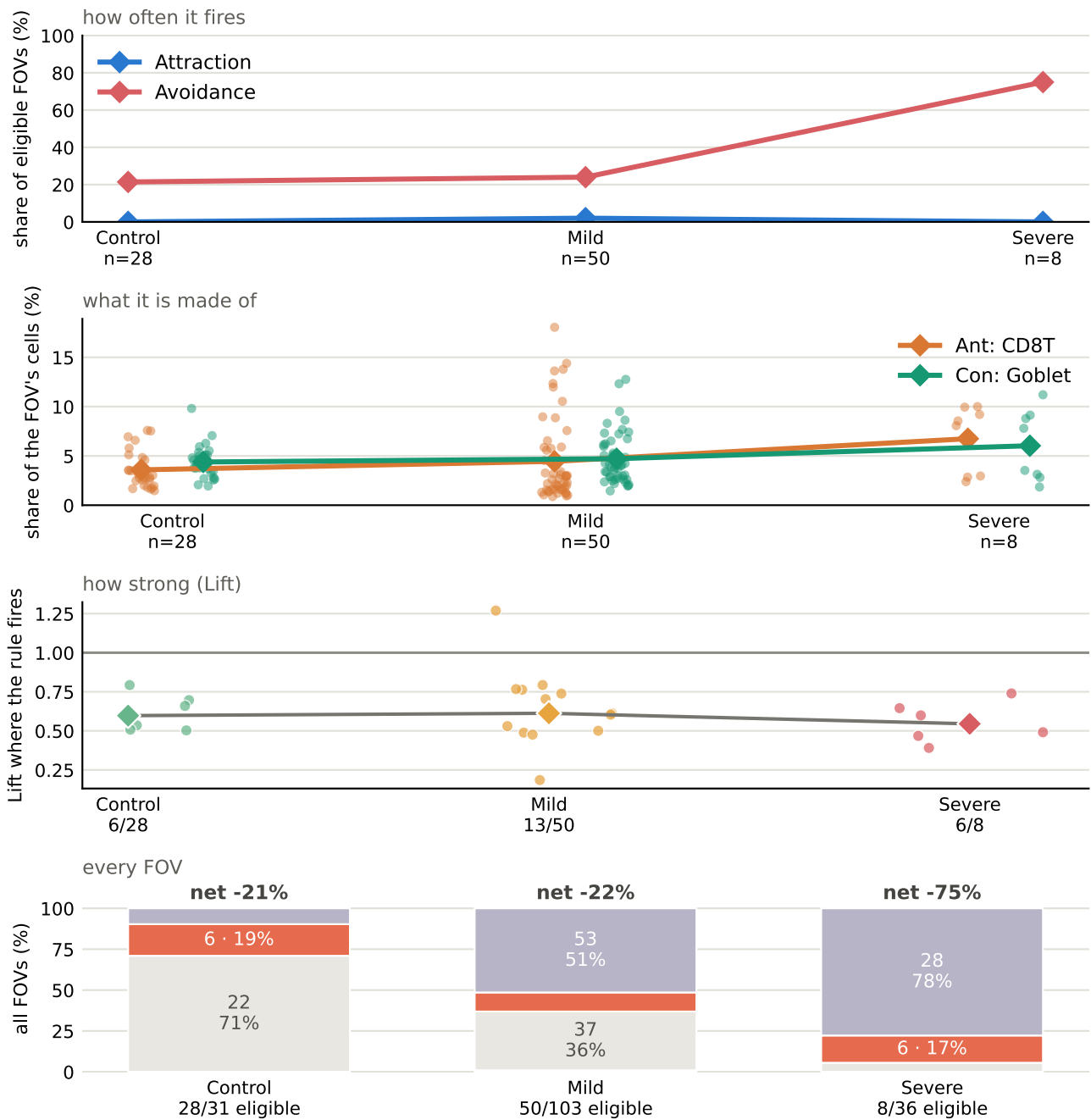


Figure 26: Values follow Figure 8; pathological stages.

Clinical staging gives Severe 17 eligible fields instead of 8, and the direction repeats – avoidance 21%, 22%, 53% – but C-S FDR is 0.315 and nothing passes. The effect is real in direction and not established in size.

Duodenum · CD8T → Goblet

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.463 · C-M FDR 0.993 · C-S FDR 0.315 · M-S FDR 0.277

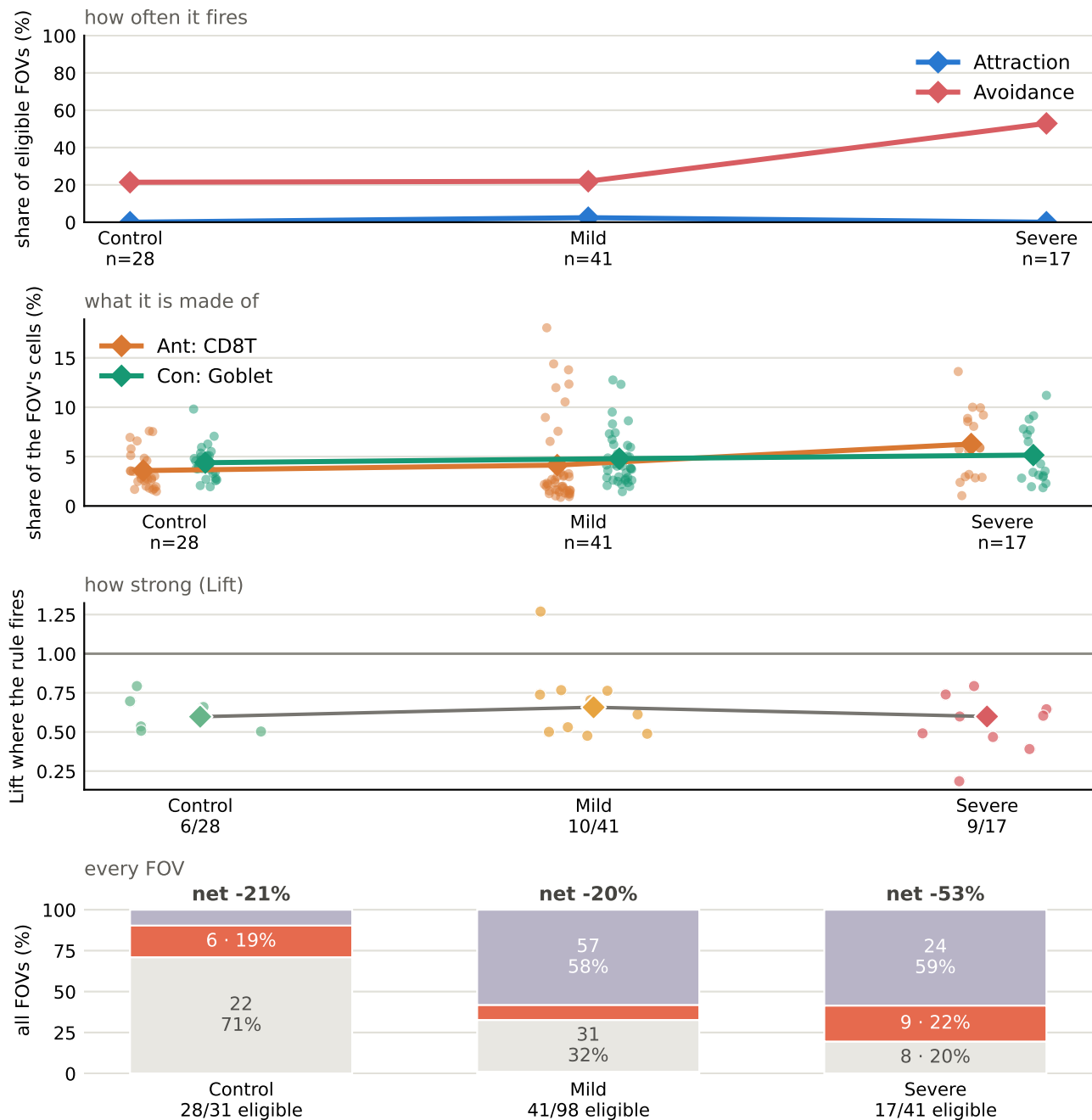


Figure 27: Values follow Figure 8; clinical stages.

The plasma-cell arrangement drops at once and never returns

The paper's temporal result is specific: lymphocytes fall sharply in the first month, myeloid cells do not, and plasma cells alone fail to recover. Grouping the same colonic fields by time since transplant reproduces that shape at the level of arrangement – CD4T → Plasma and CD8T → Plasma fall from about +55 and +40 to zero within 30 days and stay there past 100 days. Nothing here passes correction, so this is agreement in direction with an independently established result, not independent evidence.

From the atlas paper: "In contrast, plasma cells, which also dropped shortly after transplantation, failed to return to normal amounts months after transplantation."

Biopsy-level temporal pair-rule profiles — Colon

Context: Selection: paper-motivated immune rules · Analysis: pairwise rules · Unit: biopsy · Groups: Control / <30 / 30-100 / >100 · FOVs: eligible only

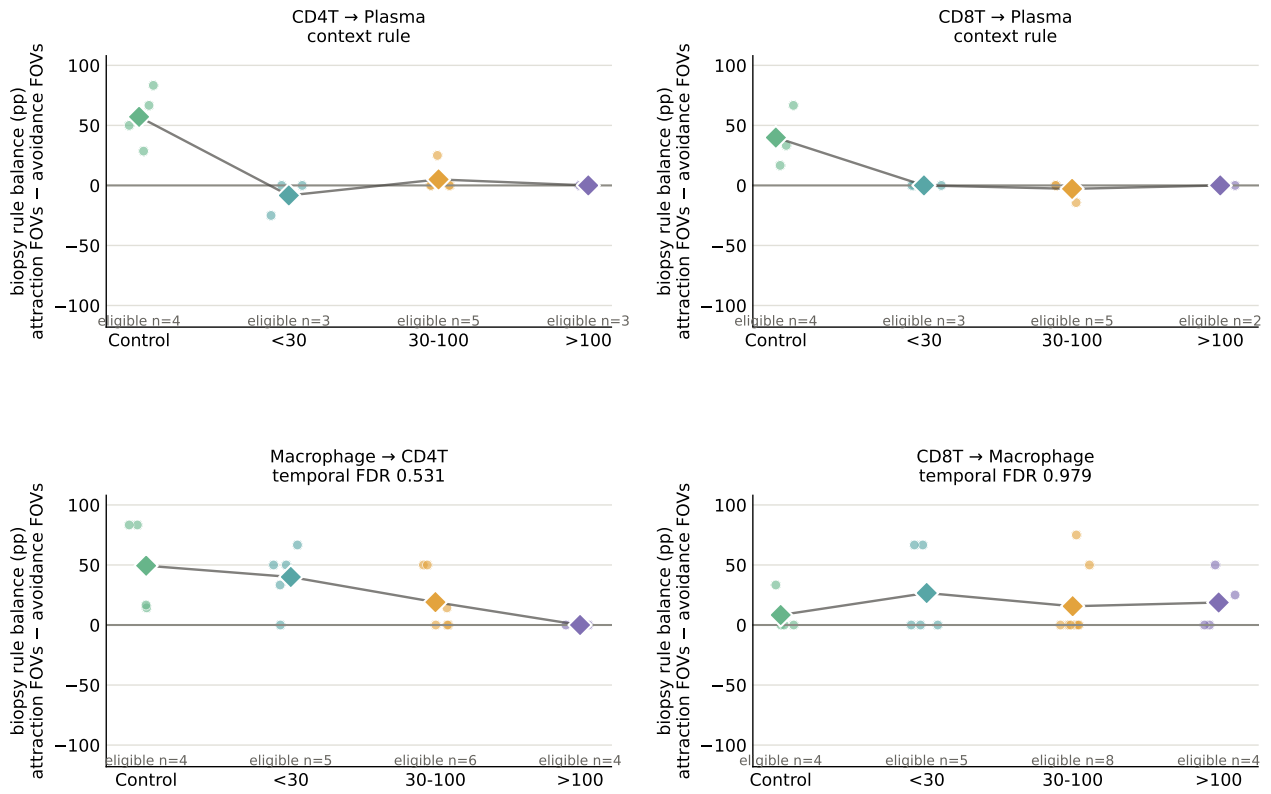


Figure 28: Each small dot is one biopsy and each diamond the group mean; the value is the share of that biopsy's eligible FOVs holding attraction minus the share holding avoidance. Control sits on the left for context and is not part of the test. The eligible biopsy count is printed under each window. These four rules were chosen because the paper highlights these immune populations, not because they ranked highly in a screen.

The same four rules stay flat in the duodenum, at temporal FDR 0.99. Both organs are shown so the contrast is visible rather than selected.

Biopsy-level temporal pair-rule profiles — Duodenum

Context: Selection: paper-motivated immune rules · Analysis: pairwise rules · Unit: biopsy · Groups: Control / <30 / 30-100 / >100 · FOVs: eligible only

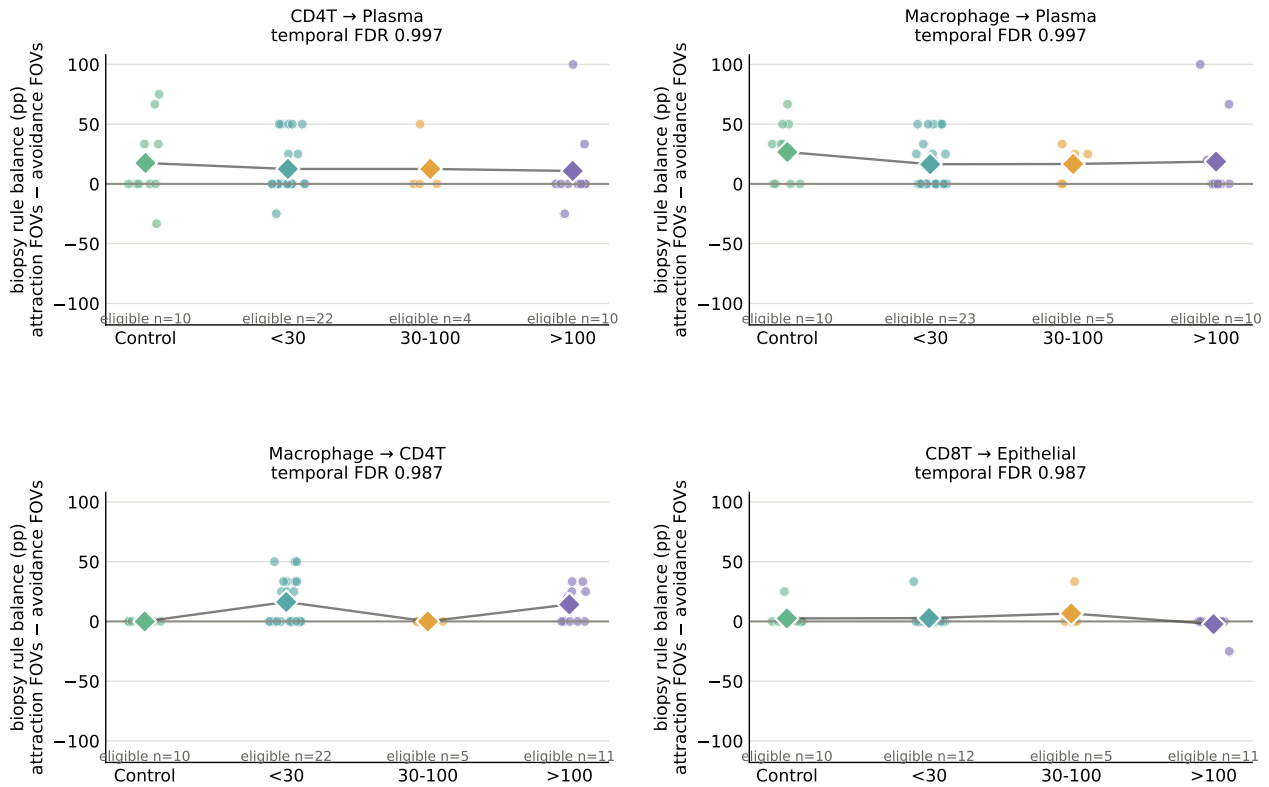


Figure 29: Values follow Figure 28; Duodenum.

3 What is new

None of the results below is stated by the atlas paper. They are ordered by how well they survive scrutiny, strongest first, because the honest headline of this analysis is how few results survive both severity scores.

Two rules survive everything

Of 525 pair rules, exactly two pass the corrected ordered trend test with eligibility applied under *both* severity scores. Both are duodenal.

CD4T → Goblet	clinical +28 pp per stage (FDR 0.022)	pathological +29 pp per stage (FDR 0.010)
Fibroblast → Muscle	clinical -29 pp per stage (FDR 0.022)	pathological -28 pp per stage (FDR 0.010)

CD4 T cells and goblet cells start almost completely separated in healthy duodenum and end barely separated at all: avoidance 77%, 53%, 19% of eligible fields. Why this is interesting: the paper shows CD4 T cells are zoned along the crypt-villus axis in healthy gut and fall in number in disease. Here their number is not what changes most – their access to the goblet compartment is. CD4T falls from 8.5% to 3% of cells while the rule moves three times as far, and the denominators are the healthiest in the document at 30, 75 and 21 fields.

Duodenum · CD4T → Goblet

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.0217 · C-M FDR 0.404 · C-S FDR 0.0557 · M-S FDR 0.167

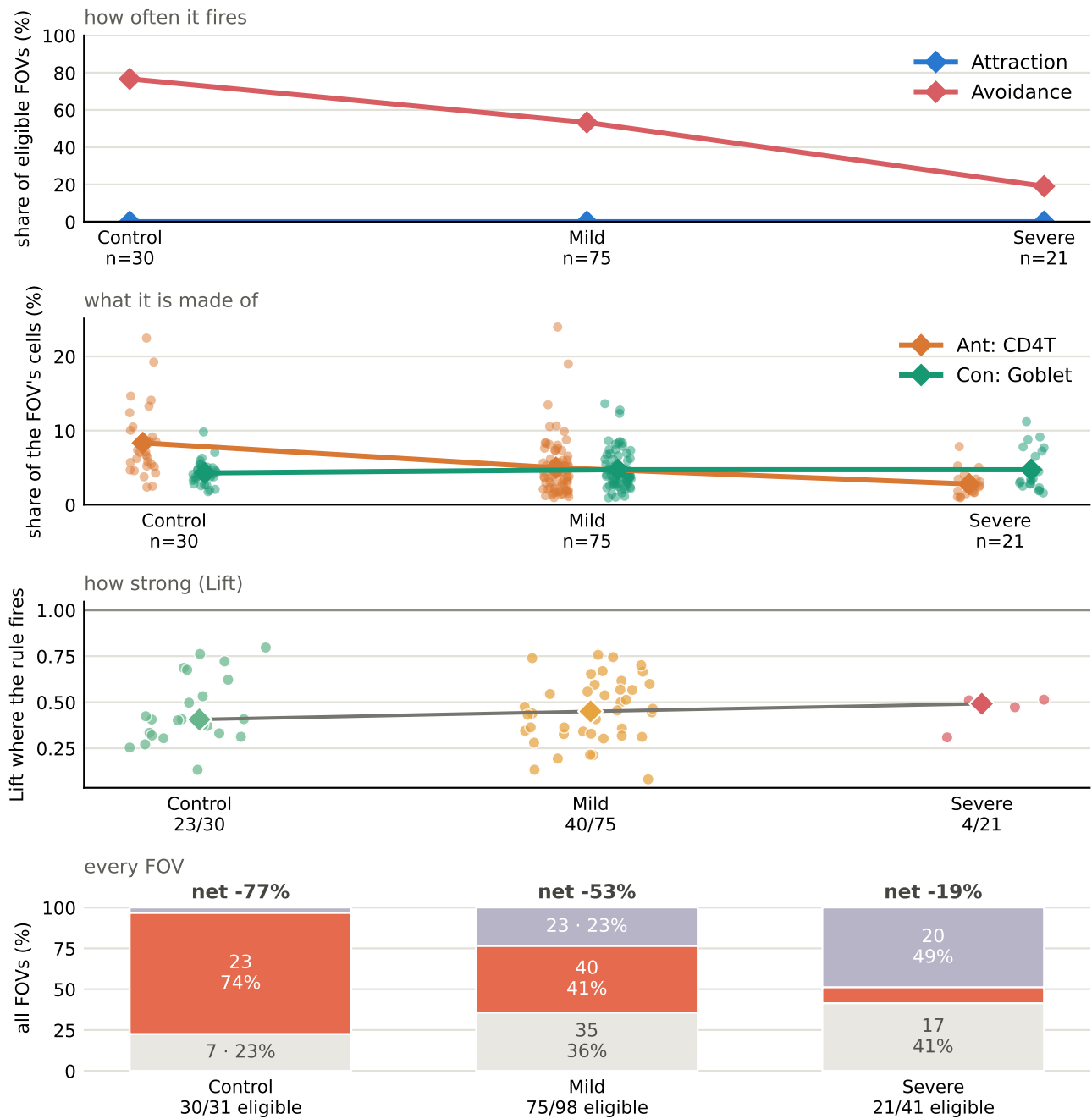


Figure 30: Values follow Figure 8.

In tissue both populations are sparse and interleaved, and the change between stages is statistical rather than visually dramatic – which is what a shift of 0.09 in Lift should look like.

Duodenum CD4T → Goblet — representative FOVs

Unit: FOV · Organ: Duodenum · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · Selection: median-strength rule-bearing FOV in each stage

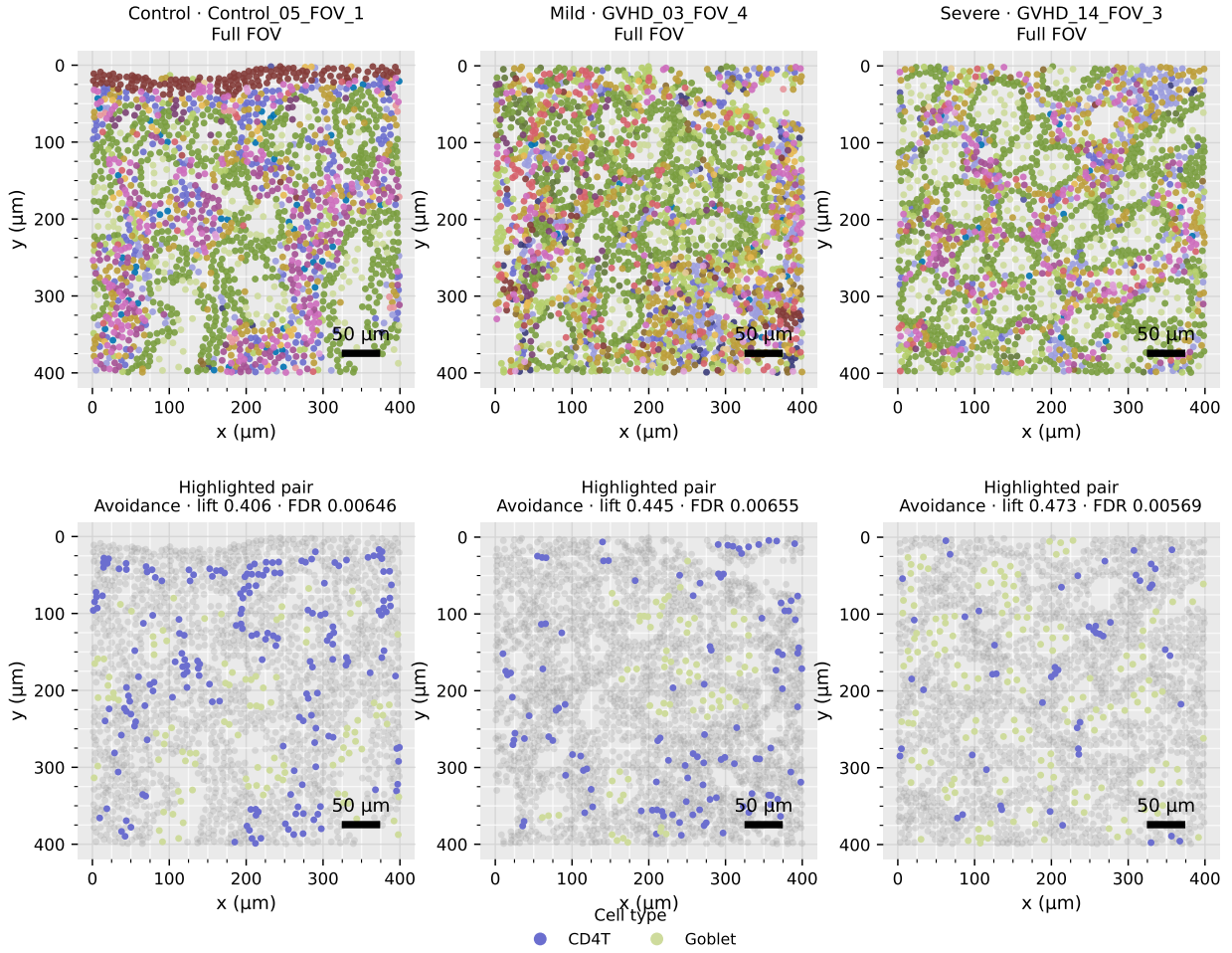


Figure 31: Selection and rows follow Figure 22.

Fibroblast → Muscle was shown in Part 2 (Figures 21 and 22) because it also matches the paper's fibrosis result. It is worth restating why it is unusual: fibroblasts and muscle both grow substantially – 8% to 14% and 4% to 20% of cells – and separate anyway. Abundance and arrangement move in opposite directions, which is the cleanest demonstration available that the two are different measurements.

Immune and stromal cells pull away from duodenal muscle

Under pathological staging the muscularis is where the duodenal signal concentrates. Macrophage → Muscle goes from 22% to 15% to 70% avoidance (ordered FDR 0.010, C-S 0.038) on 18, 66 and 27 eligible fields – healthy denominators for this cohort – while both cell shares roughly double and the unjudgeable block shrinks from 42% to 25%. The Control-to-Severe gap is what carries it; the dip at Mild means the path there is not a clean gradient.

Why this is interesting: the muscularis has its own resident macrophage population that sits close to the muscle and to enteric neurons and helps regulate gut motility. That is established gut biology, independent of transplantation. Losing the macrophage–muscle association is therefore a candidate readout for muscularis disruption, and it sits alongside the paper's report of increased fibrosis in the same tissue. What the rules cannot do is identify the macrophage subtype, so this is a lead about a compartment, not about a cell state.

Duodenum · Macrophage → Muscle

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.0101 · C-M FDR 0.832 · C-S FDR 0.0378 · M-S FDR 0.0132

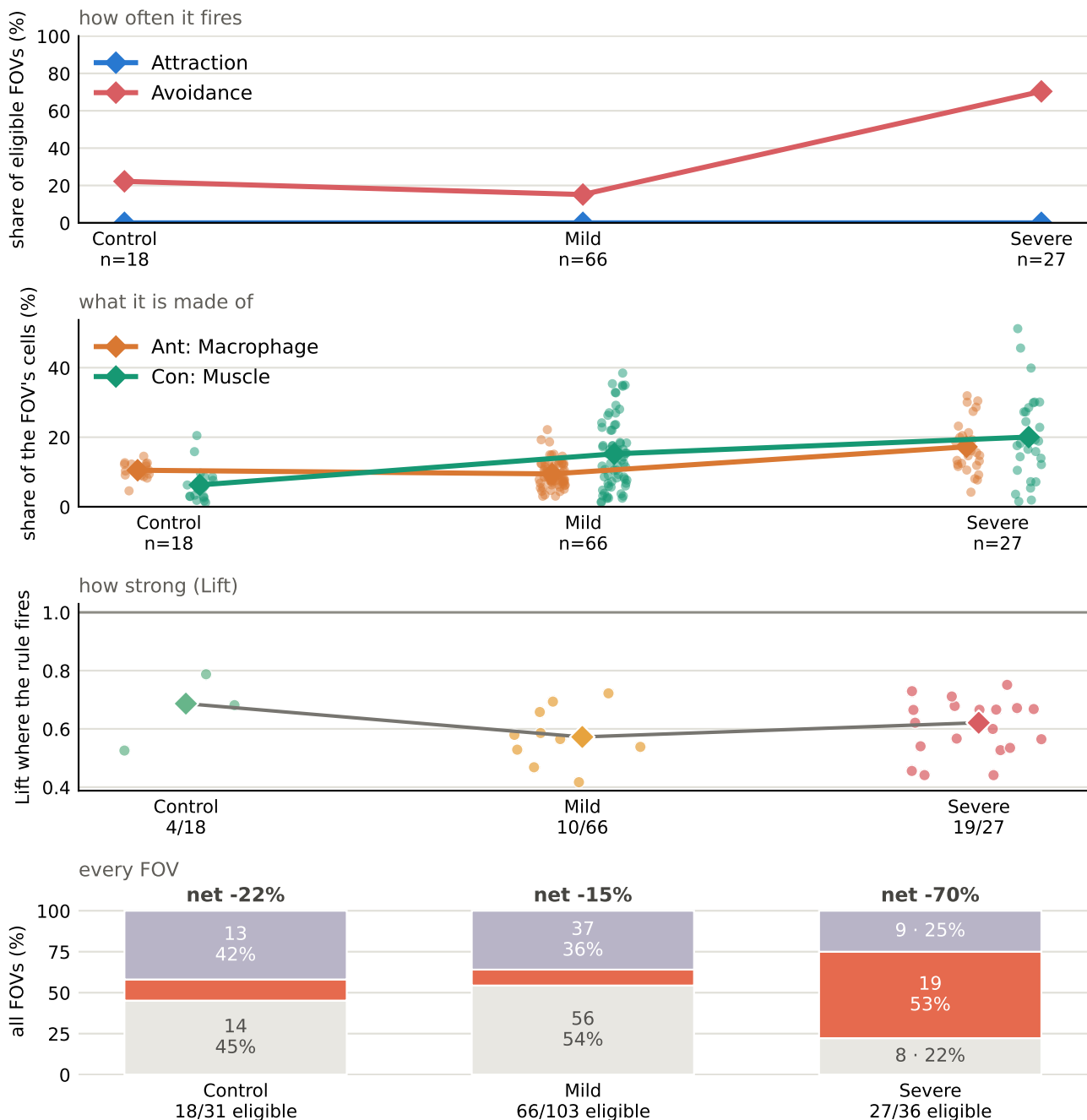


Figure 32: Values follow Figure 8; pathological stages.

The Paneth niche reorganises, on very few fields

Read as a whole rather than rule by rule, the pathological duodenal screen concentrates on one compartment: six of the rules it selects name Paneth cells – Paneth → Mast, Paneth → Macrophage, Paneth → Plasma, Paneth → Plasma_CD38, CD8T → Paneth and Epithelial → Paneth. Paneth cells are well under 1% of duodenal cells, so nothing about their abundance would put them at the top of a screen.

Why this is interesting: Paneth cells sit at the base of the crypt beside the intestinal stem cells and are the antimicrobial anchor of that niche. The paper proposes that stem cells in aGVHD may misdifferentiate into endocrine cells instead of Paneth cells. If the crypt-base niche is being rebuilt, the contacts that define it should change – and those contacts are what these rules measure. The rule layer is pointing at the compartment the paper singles out, through the neighbours rather than the counts.

The caveat comes first and is severe: these rules rest on 3 to 6 eligible Severe fields, because Paneth cells are rare

and become rarer, and none of them appears at all under clinical staging. Paneth → Mast reaches ordered FDR 0.010 on 9, 38 and 3 eligible fields – so its 67% Severe avoidance is 2 fields out of 3. This is a lead, not a result.

Duodenum · Paneth → Mast

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.0101 · M-S FDR 0.0132

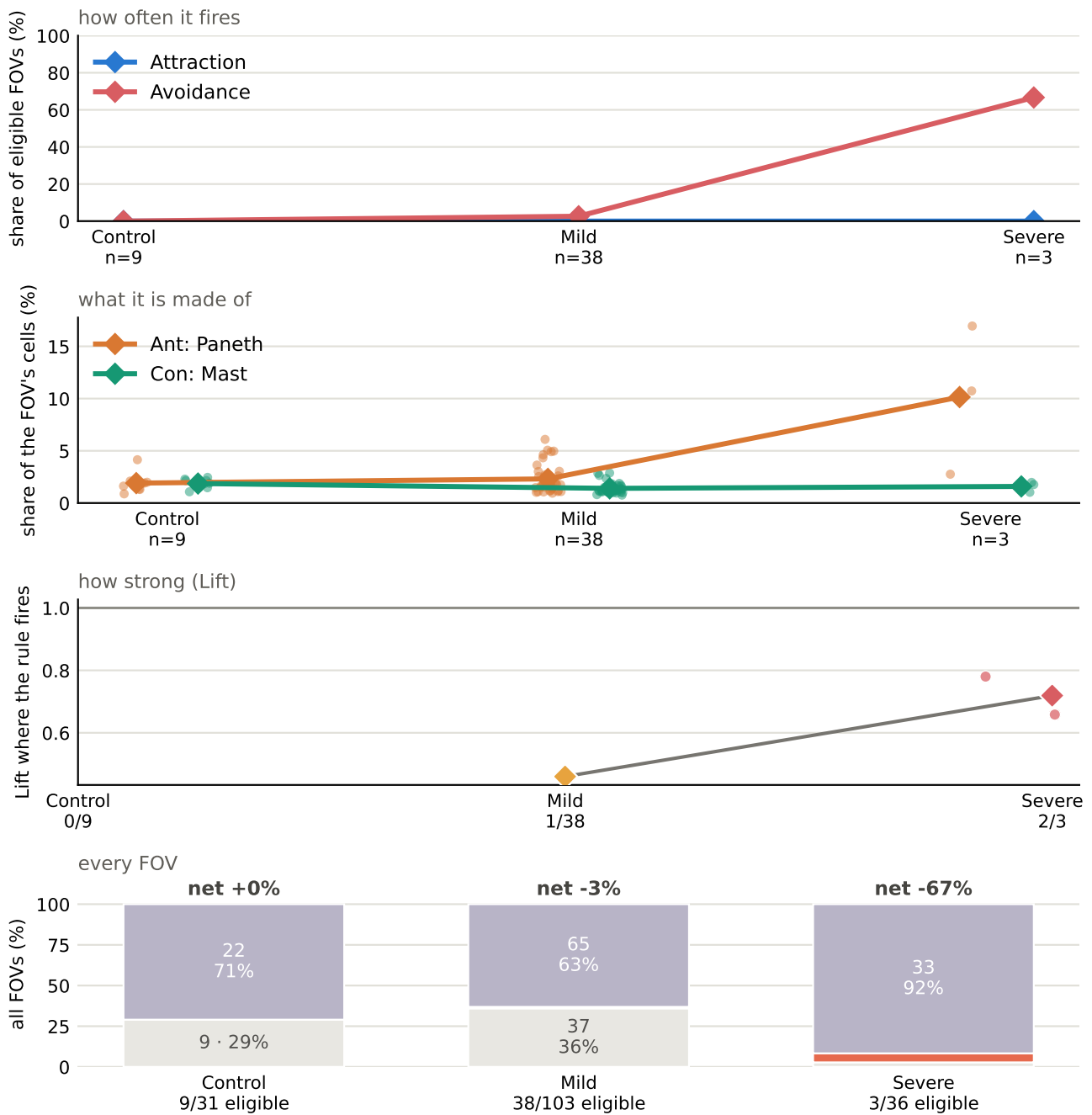


Figure 33: Values follow Figure 8; pathological stages.

Paneth → Macrophage adds something the others do not: avoidance both spreads (8%, 21%, 33%) and deepens (Lift 1.2 to 0.58), at ordered FDR 0.046 on 13, 63 and 6 fields.

Duodenum · Paneth → Macrophage

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
 Tests: ordered FDR 0.0457 · C-M FDR 0.135 · C-S FDR 0.135 · M-S FDR 0.655

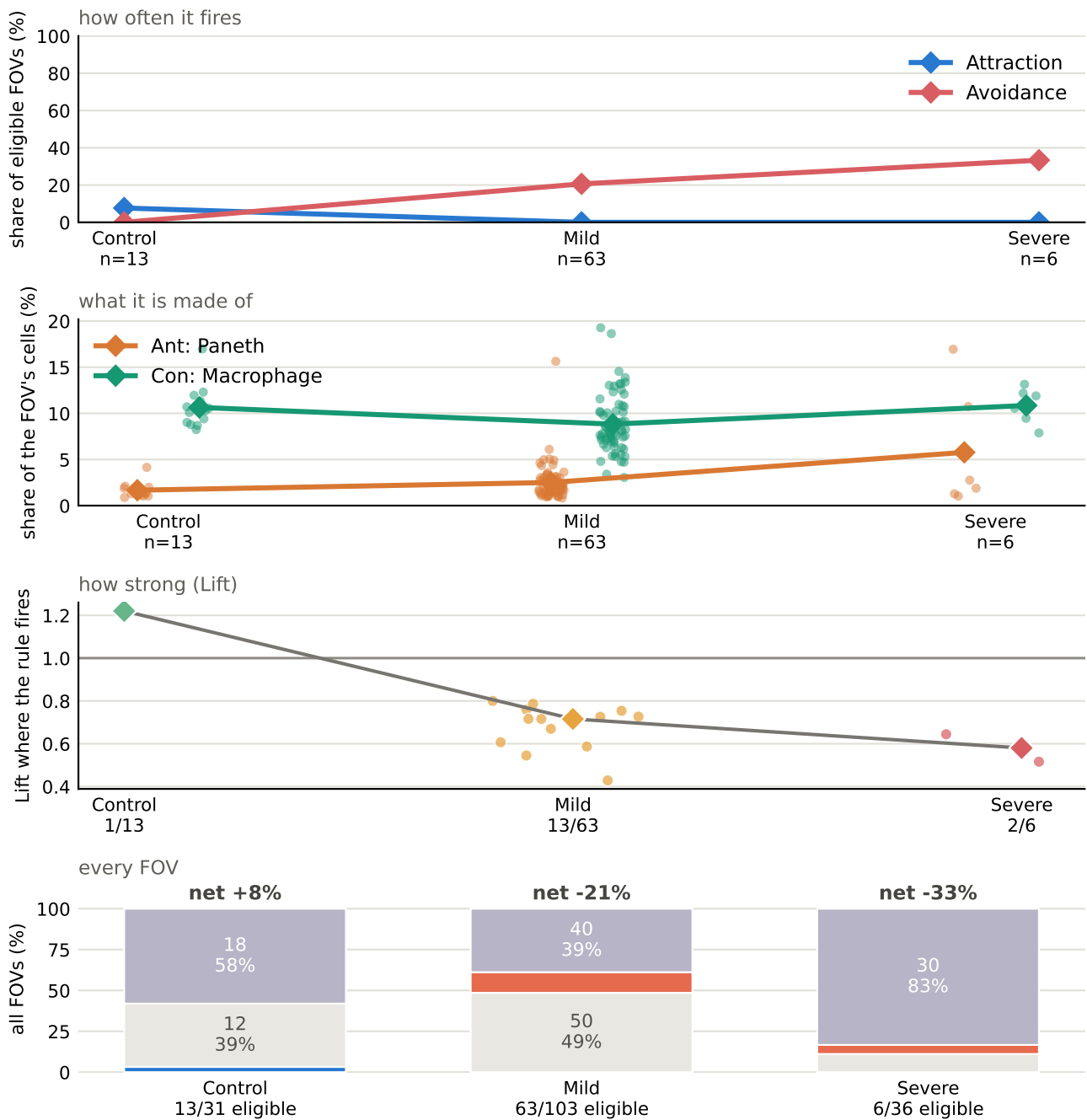


Figure 34: Values follow Figure 33.

CD8 T cells are the effector population of the disease, so their relation to the crypt base is the most mechanistically interesting member of the family and the one most worth re-testing in a larger cohort – on 12, 38 and 4 eligible fields, M-S FDR 0.025.

Duodenum · CD8T → Paneth

Unit: FOV · Organ: Duodenum · Score: pathological · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.0677 · M-S FDR 0.0245

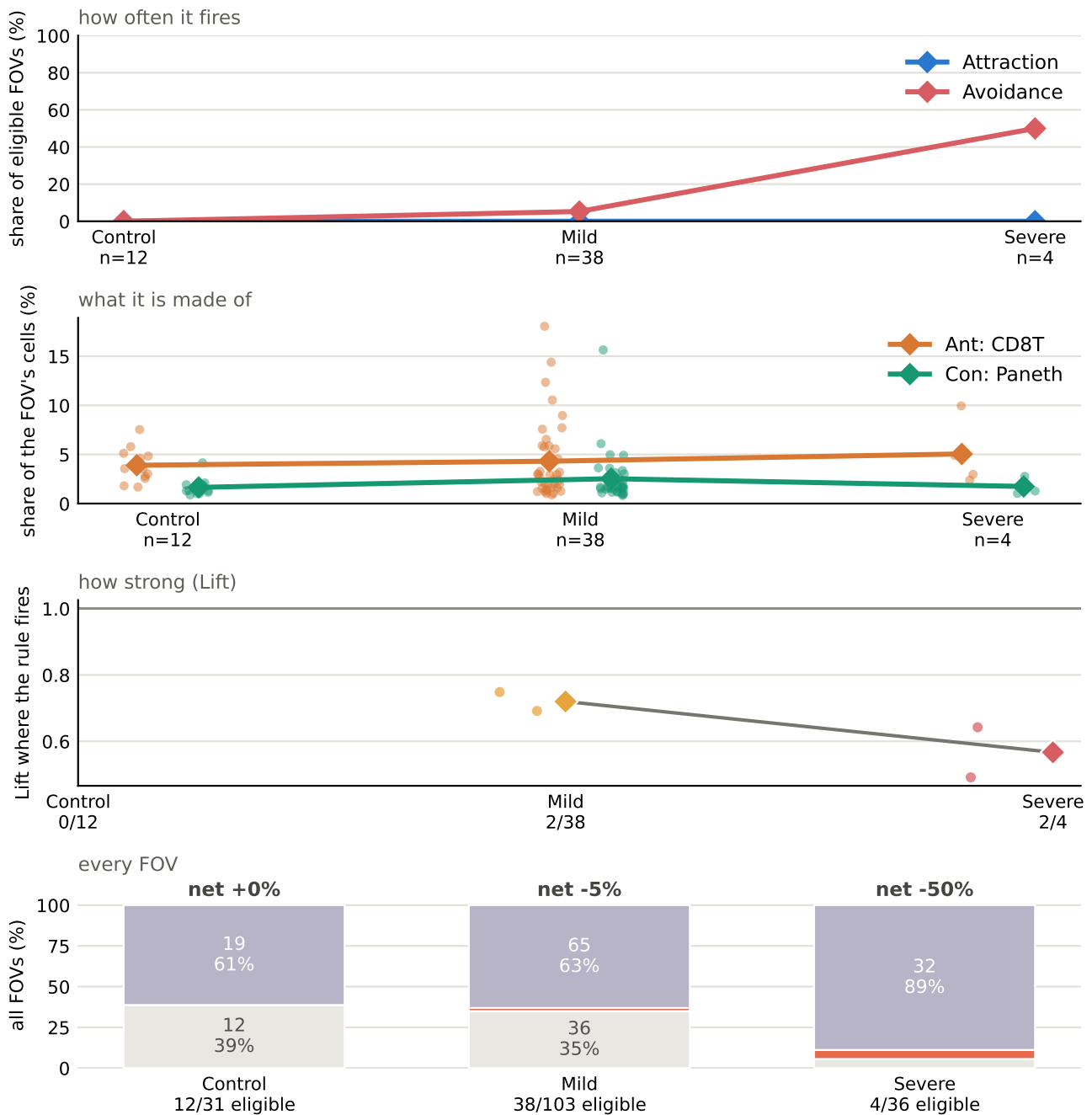


Figure 35: Values follow Figure 33.

Plasma cells and blood vessels swap places in the colon

Colon Plasma → SMV inverts: net -10% , $+25\%$, $+24\%$, with the median Lift crossing from 0.69 to 1.35 (C-M FDR 0.029). Why this is interesting: plasma cells normally sit in the lamina propria above the crypts, and a switch from avoiding small vessels to associating with them is what a population arriving from the circulation rather than residing in the tissue would look like – the transition the paper’s donor-host chimerism work describes. The Mild group holds 4 eligible fields, so the middle of the trajectory is thin.

Colon · Plasma → SMV

Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type · n = eligible FOVs
Tests: ordered FDR 0.175 · C-M FDR 0.029 · C-S FDR 0.128 · M-S FDR 1

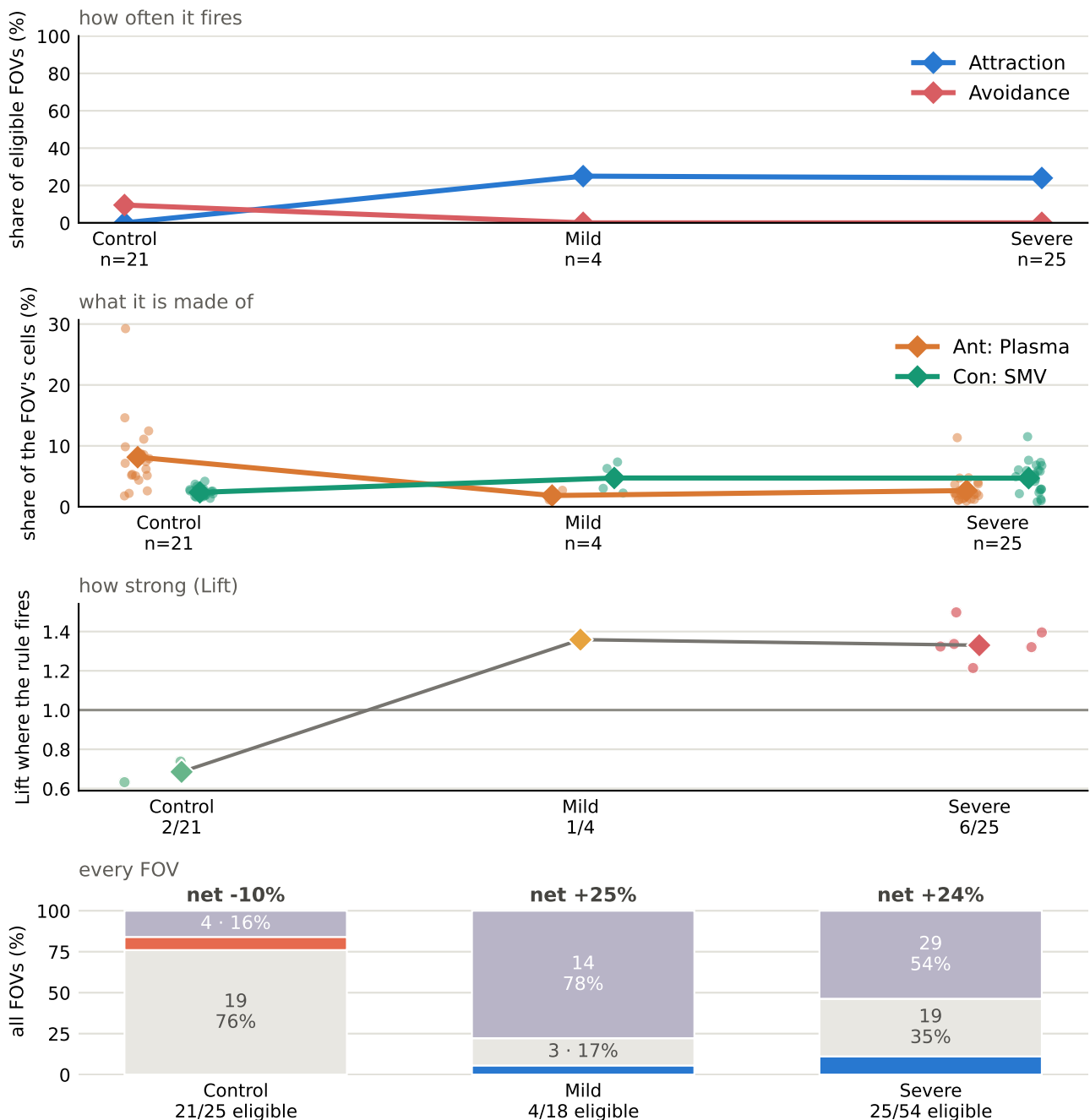


Figure 36: Values follow Figure 8; clinical stages.

How strong a rule is, versus how often it is found

Everything so far counted how often a rule fires. That misses a whole class of change: a rule found just as often but much stronger or weaker where it is found. Prevalence and Lift are therefore screened separately, each ranked by its largest stage gap, with no rule named in advance. The two screens do not return the same rules, which is the point.

Colon Plasma → Goblet is the clearest case. Its avoidance Lift drops from 0.73 to 0.22 – the two types end up roughly three times further apart than chance would place them – at FDR 0.008, while its prevalence hardly moves at all, at FDR 0.68. A prevalence-only analysis would have recorded this rule as unchanged.

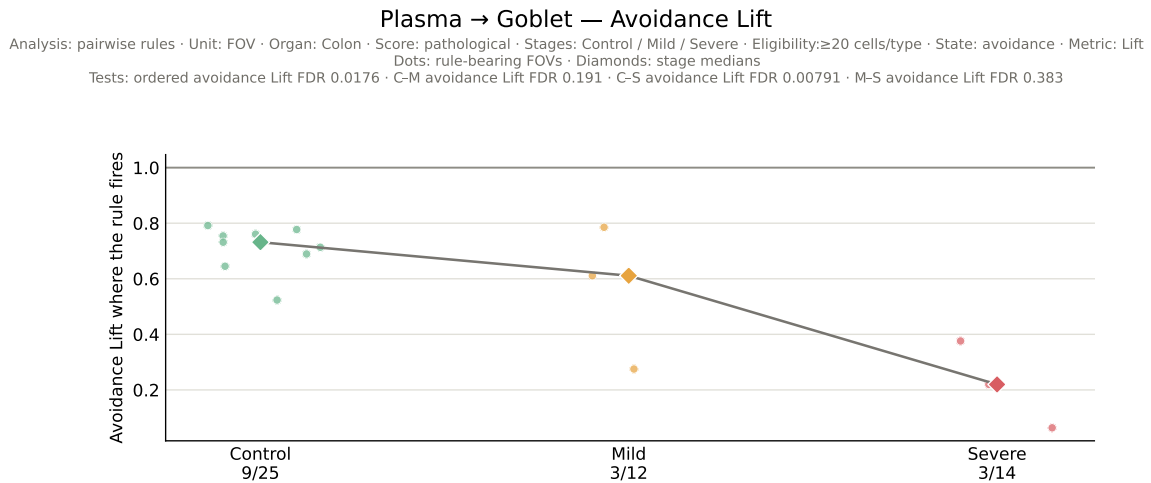


Figure 37: Each dot is one rule-bearing FOV, restricted to occurrences of a single direction, with the stage median as a diamond. For avoidance, lower Lift means stronger separation. The count below is rule-bearing / eligible FOVs. Colon, pathological stages, avoidance occurrences only. The Severe median rests on 3 rule-bearing fields, so the size of the drop is less certain than its direction.

Scored by prevalence instead, the very same rule looks unchanged.

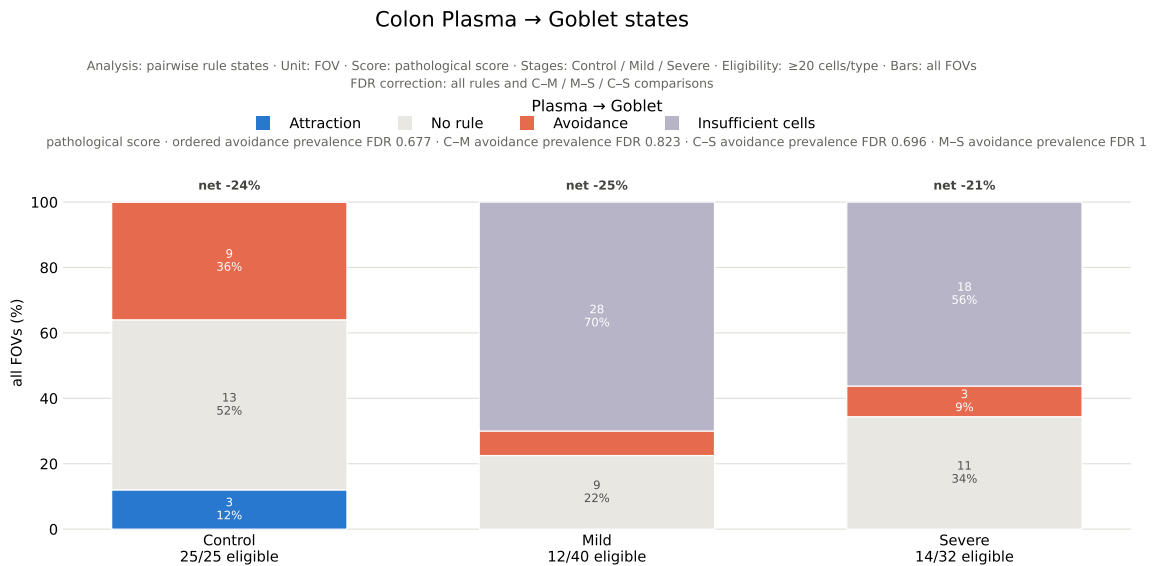


Figure 38: The same rule scored by prevalence instead of strength, drawn as the bottom panel of Figure 8. Read against Figure 37: two different corrected screens reaching opposite conclusions about one rule.

The attraction side has a matching case with a clear biological reading. Colon SMV → Endothelial keeps firing in most eligible fields – net +73%, +33%, +74%, ordered prevalence FDR 0.978 – but weakens, median Lift 3.45 to 2.37 at C-S FDR 0.065, and reaches ordered FDR 0.015 under pathological staging. Why this is interesting: damage to the endothelium is a recognised feature of transplant complications, and vessels are one of the few compartments where a loosening of structure has an obvious physical meaning. This rule measures the association between vessel-lining cells and the small and medium vessels they belong to, so a weakening association is a plausible spatial signature of vascular disruption. The atlas does not report it.

SMV → Endothelial — Attraction Lift
 Analysis: pairwise rules · Unit: FOV · Organ: Colon · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥20 cells/type · State: attraction · Metric: Lift
 Dots: rule-bearing FOVs · Diamonds: stage medians
 Tests: C-S attraction Lift FDR 0.0648

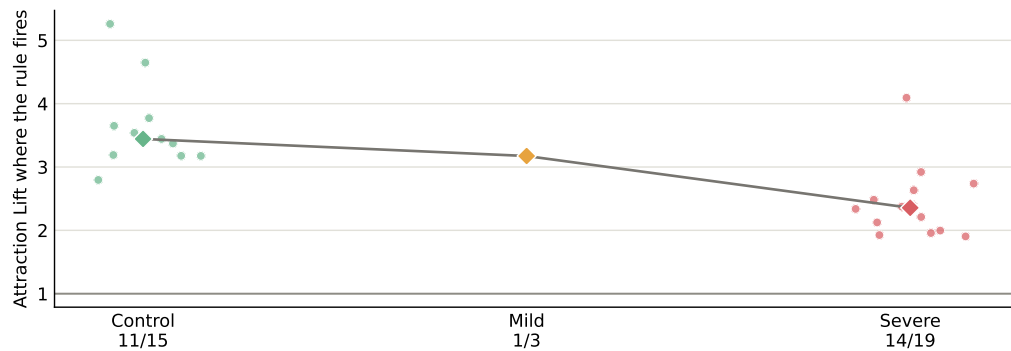


Figure 39: Values follow Figure 37, for attraction occurrences, where higher Lift means stronger association. Clinical stages.

Its prevalence, by contrast, is as high in Severe disease as in Control.

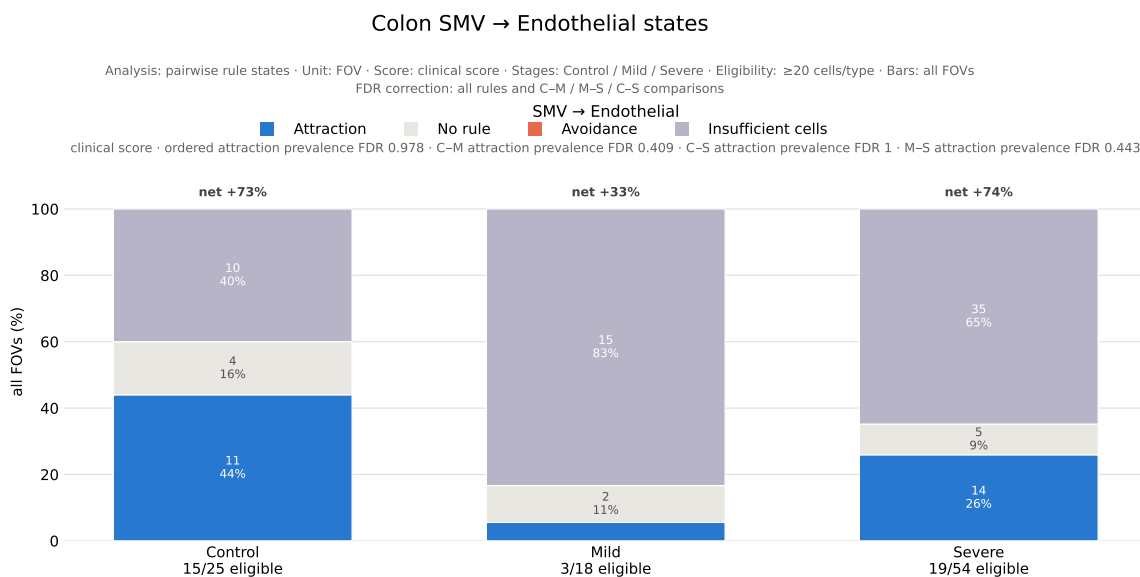


Figure 40: Prevalence for the same rule, drawn as the bottom panel of Figure 8.

Time after transplant: an honest negative

The paper makes a strong case that time since transplantation must be modelled, so it is worth stating what happens when the rules are grouped that way instead of by severity: at biopsy level, nothing survives correction. In both organs zero rules pass FDR 0.05 in the three-window screen, and zero pass in either pooled contrast.

The reason is visible in the sampling. The three windows are unevenly filled and the “more than 100 days” group spans everything from 100 to over 1100 days, so it is not a homogeneous group at all.

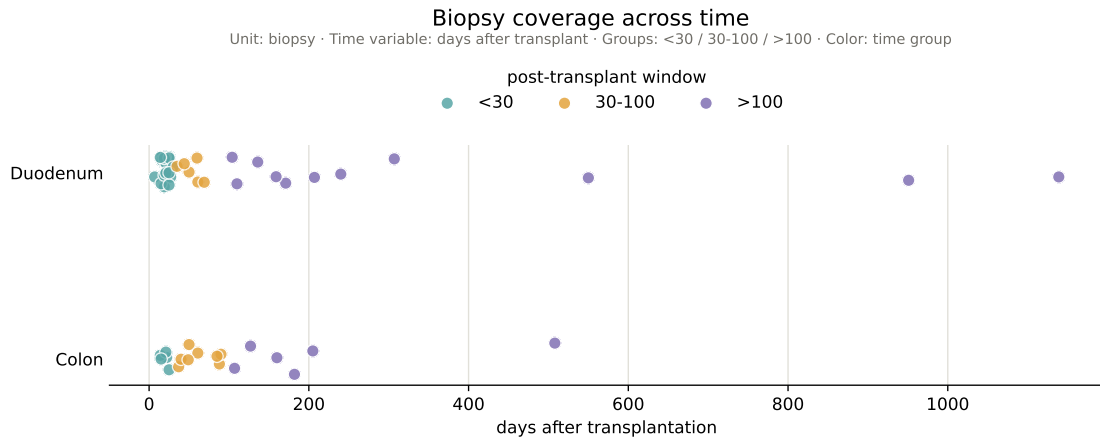


Figure 41: Each dot is one biopsy at the number of days between transplantation and biopsy, coloured by the window it was assigned to. Biopsies are the unit of every temporal test, so this is the effective sample size of this whole subsection.

Screening every rule against the three windows returns nothing at all in the colon.

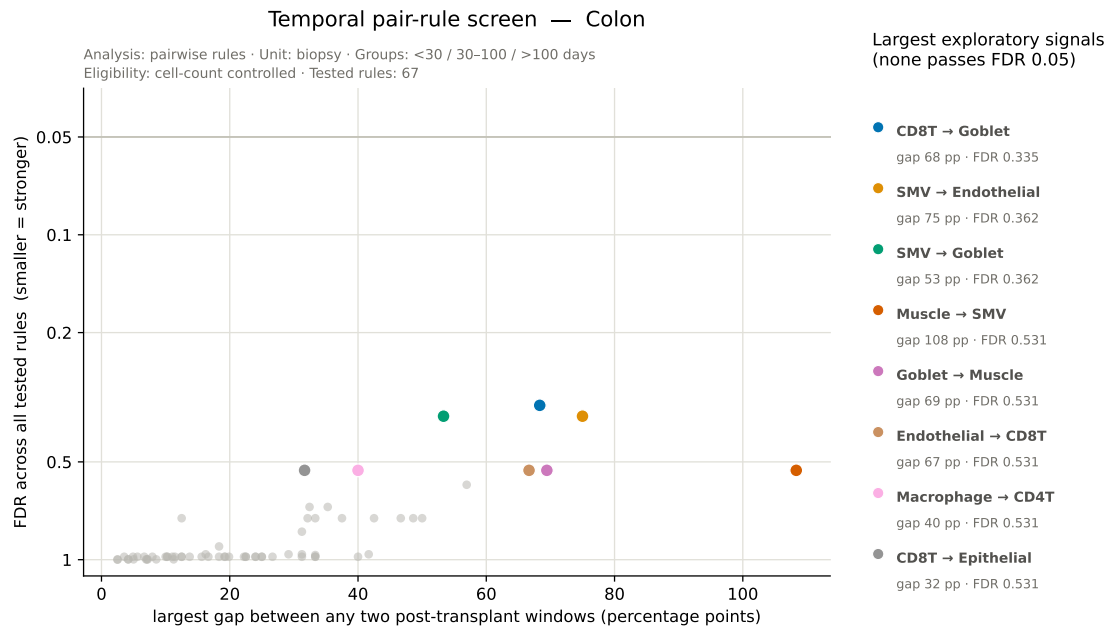


Figure 42: Each dot is one rule: the horizontal axis is the largest gap between any two post-transplant windows, the vertical axis the FDR across all rules tested at biopsy level, drawn so stronger results sit higher. Colon.

The largest duodenal signals are all plasma-cell and goblet-cell rules, which is at least the compartment the paper's kinetics point to, but none of them passes either.

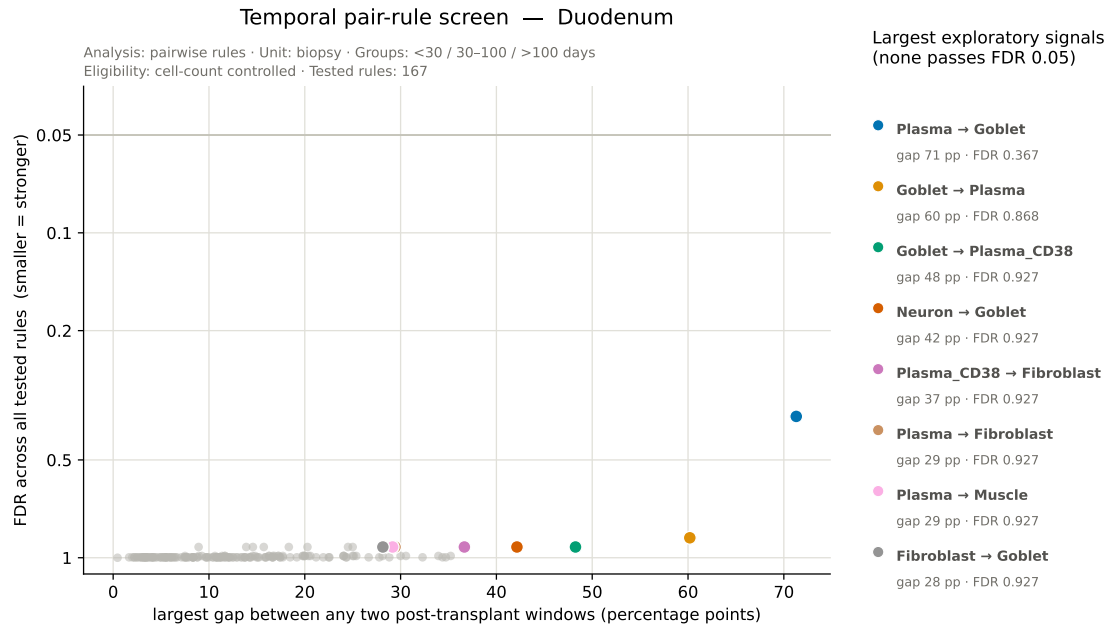


Figure 43: Values follow Figure 42; Duodenum.

Pooling the windows into two broader comparisons does not rescue it.

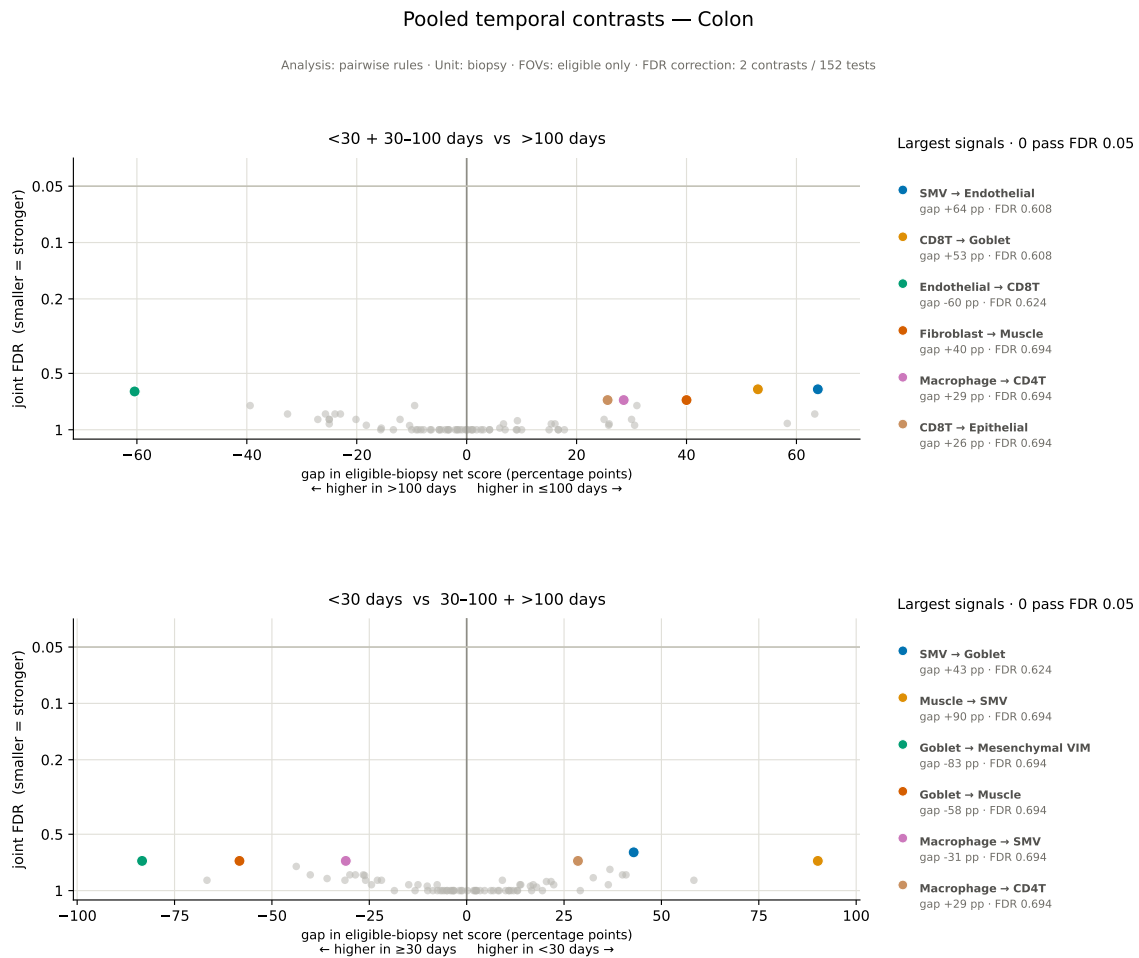


Figure 44: Two contrasts, each read as in Figure 42, with one FDR correction applied jointly across both contrasts and all their rules.

4 What to take from this

The method measures arrangement, not composition. Across every comparison in Part 1 the rules move several times further than the cell counts behind them – 67 percentage points against 9 in the clearest case – and Fibroblast → Muscle separates while both of its cell types grow.

Eligibility control is not a detail, it is most of the analysis. Without it, 22 colonic rules climb and 11 fade with clinical severity. With it, none of them passes. A rule going from avoidance to unjudgeable looks exactly like avoidance relaxing, and in the colon – where goblet cells fall from 22% to 13% of the tissue – that is what most of the apparent signal was.

Where the atlas has a finding, the rule layer recovers it independently: Paneth loss, endocrine accumulation, the collapse of the colonic plasma-cell network, the stromal shift, the organ split, and the immediate and permanent loss of plasma-cell organisation after transplant.

Two results survive everything. Duodenal CD4T → Goblet and Fibroblast → Muscle pass the corrected ordered trend test with eligibility applied under both severity scores, with the best denominators in the cohort. Everything else in Part 3 is score-dependent: the Paneth family and the monotonic colonic goblet gradient are pathological findings, the colonic plasma attractions are clinical findings.

That score-dependence is itself the most interesting thing here. It is not noise. The epithelial results track the pathological score, which is read off the same slide the rules are measured in; the immune results track the clinical score, which is the one the paper uses for its own immune work. The paper notes that the two gradings are only moderately correlated and suggests imaging features might bridge them. These rules behave exactly as though the two scores are measuring different things.

The limits. Many endpoints rest on a handful of eligible fields – the Paneth family worst, at 3 to 6. The temporal analysis is a clean negative at biopsy level. And every result here is a spatial association between classifier labels, which is where a mechanism starts rather than evidence of one.